Firm Scope and Innovation: The Role of Intangibles*

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Abstract

This paper develops a unified theoretical framework to study how the composition of intangible investment and innovation direction jointly shape long-run growth and creative destruction. I build an endogenous growth model in which firms choose between vertical and horizontal innovation and allocate investment across R&D and firm-specific intangibles (brand value and organizational capital). Using firm-level data, I discipline the model and find that productivity, markups, and intangible investment composition vary systematically with firm scope: firms with narrower scope concentrate investment in knowledge capital, whereas firms with broader scope allocate relatively more to brand value and organizational capital. The calibrated model shows that firm scope determines intangible composition, which governs innovation direction, and that growing scope and organizational capital jointly raise entry barriers and suppress creative destruction. Size-based policies misrepresent firms' investment incentives by treating vertical and horizontal expansion symmetrically, whereas scope-dependent policies targeting intangible-driven externalities lead to higher rates of creative destruction and innovation.

Keywords: Schumpeterian growth, step-by-step innovation, intangibles, firm dynamics, span of control.

JEL Classification: E22, O31, O32, O33, O34.

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1. Introduction

Horizontal and vertical innovation are two core strategies in the modern endogenous growth literature for generating long-run growth through creative destruction. Firms can engage in horizontal innovation by expanding their product portfolios (Klette and Kortum, 2004) or pursue vertical innovation by improving the quality of existing products (Aghion and Howitt, 1992; Aghion *et al.*, 2001). At the same time, literature also documents that firms invest in intangibles not only to innovate but also to build intangible assets such as brand value and organizational capital that strengthen market positions rather than drive innovation-led growth (Cavenaile and Roldan-Blanco, 2021; Crouzet and Eberly, 2019).

While the directions of innovation and composition of intangibles are well documented separately, understanding their interaction is crucial for shaping effective policies. To be concrete, Apple is a vertically integrated firm focused on mobile communication technologies and invests heavily in R&D. By contrast, 3M is active in transportation, safety, consumer, and health care industries, and although R&D remains its largest intangible investment, it allocates a substantially larger share toward brand and organizational capital relative to Apple [1]. This contrast in firm scope shows weakness of flat and size-based policies. By treating Apple and 3M symmetrically, such policies can either penalize firms that invest heavily in R&D or equally reward those with excessively accumulated firm-specific intangibles, regardless of their aggregate consequences. Yet existing theories provide no framework for distinguishing these two firms' investment incentives or for designing policies that account for this heterogeneity.

What are the aggregate consequences of the interaction between innovation direction and intangible composition for economic growth and creative destruction? To answer this question, I proceed in two steps. First, I develop a unified theoretical framework in which firms engage in both vertical and horizontal innovation and allocate investment across different types of intangibles. Second, I document a new set of empirical regular-

[1] The measurement details are given in Section 3.2.

ities showing how firm-level outcomes and investment choices vary systematically with firm scope, which I use to discipline the model. The calibrated model shows that firms operating with a narrower scope concentrate investment in knowledge capital, whereas firms with a broader scope allocate relatively more to brand value and organizational capital, and the composition of intangible types in turn determines firms' innovation direction. This joint determination of innovation direction and intangible composition implies that policies conditioned solely on firm size are insufficient, as firm size contains vertical and horizontal dimensions and therefore misrepresents firms' intangible investment incentives. By contrast, scope-dependent policies that target the externalities associated with intangible heterogeneity lead to higher rates of creative destruction and innovation.

To formalize intangible heterogeneity, I classify intangible assets according to their spillover effects across firms. In this paper, transferable intangibles consist of knowledge capital[2] with the associated R&D investments. These assets can be transferred between firms and generate spillover effects due to their non-rivalrous and partially excludable nature (Romer, 1990). In contrast, embedded intangibles comprise brand value[3] and organizational capital[4] are inherently firm-specific and inseparable from the firm that created them and do not generate spillovers. Consequently, when a firm exits the market, the economic value of embedded intangibles becomes a sunk cost.

In Section 2, I develop an endogenous growth model that the economy consists of a single final-good sector and a continuum of intermediate-good sectors. In each intermediate goods sector, a single superstar firm and a continuum of fringe firms produce intermediate goods for the final goods sector with a la Bertrand competition. Superstar

[2]Following Griliches (1979), R&D investment generates a stock of knowledge capital that accumulates over time. Examples of such knowledge include patents and innovation-related software, which illustrate the types of assets that represent technological know-how.

[3]Brand value is a demand shifter, positively influencing the perceived quality of a firm's output (Cavenaile and Roldan-Blanco, 2021; Cavenaile *et al.*, 2025a). Evidence also suggests that brand value increases consumer awareness through targeted marketing (Cavenaile *et al.*, 2025b; Baslandze *et al.*, 2023)

[4]Organizational capital conceptualized as managerial productivity, including the firm's embodied managerial talent and its contribution to future production profitability (Carlin *et al.*, 2012; Eisfeldt and Papanikolaou, 2013; Prescott and Visscher, 1980)

firms can invest in embedded intangibles, which improve managerial productivity and the perceived quality of their products through brand value and organizational capital. Alternatively, they can invest in transferable intangibles, which enhance product quality in two ways: (i) by vertically improving existing products and (ii) by expanding their scope through entry into new sectors. Fringe firms produce homogeneous products and cannot accumulate brand value and organizational capital. When a superstar firm exits a sector, its transferable intangibles—the established product quality in that sector—become freely available to fringe firms. For fringe firms, the only path to becoming a superstar is through radical innovation.

The model incorporates two key frictions. First, two types of intangibles differ in their scalability. Embedded intangibles generate economies of scope, as their benefits increase when a firm is active in more sectors because they are fully mobile across sectors, whereas transferable intangibles require separate investment for each sector. Second, while vertical innovation imposes no span-of-control constraints, expanding scope divides managerial attention across sectors, therefore reducing managerial quality per sector (Lucas, 1978)[5]. However, when a superstar firm expands into more sectors, its total managerial quality increases, so that while managerial quality per sector declines with span-of-control constraint, the superstar's total managerial capacity increases with scope. Moreover, superstar firm could also strengthen their managerial quality through the accumulation of organizational capital. Therefore, when facing a competitive threat, the incumbent superstar leverages both its embedded intangibles and scope, creating entry barriers for challengers by deploying its total managerial capacity.

In Section 3, I combine firm-level data from Compustat with the product-market fluidity dataset of Hoberg *et al.* (2014) to document how markups, productivity, intangible investment, and competitive pressure vary systematically with firm scope. To isolate causal direction, I employ the patent-value dataset from (Kogan *et al.*, 2017) as an innovation shock to compare the responses of single- and multi-sector firms using local

[5]See also Jovanovic (2025). Alternatively, Acemoglu *et al.* (2018) propose that when a firm expands its scope, more skilled labor needs to be allocated to operational activities.

projections. Single-sector firms show stronger reallocation toward transferable intangibles, larger productivity gains, higher markup responses, and greater competitive pressure from rivals compared to multi-sector firms facing an equivalent shock.

In Section 4, I calibrate the model to firm-level cross-sectional moments and derive three quantitative predictions. First, scope exerts a first-order effect on innovation direction: as firms expand into additional sectors, the span-of-control constraint erodes market share in each individual sector, shifting incentives away from transferable and toward embedded intangible investment. Second, even firms of similar scope pursue sharply different innovation strategies depending on the composition of their intangible capital, generating rich nonlinear heterogeneity that standard models neglect. Third, as scope and embedded intangibles grow jointly, the incumbent's managerial capacity rises, raising the displacement threshold for entrants and suppressing creative destruction. These three forces together drive a wedge between private and social returns to innovation: broader-scope firms investing largely in embedded intangibles erect larger entry barriers and generate smaller knowledge spillovers, while focused firms concentrating in transferable capital contribute more to creative destruction and aggregate productivity growth.

In Section 5, I quantify the aggregate effects of embedded intangibles and firm scope. The results indicate that embedded intangibles account for the bulk of misallocation for superstar firms, as they allow broad-scope firms to erect entry barriers that reduce creative destruction while supporting the innovation rents that sustain long-run growth. Removing embedded intangibles reallocates incumbent investment toward horizontal expansion, increasing growth primarily through superstar firms. By contrast, relaxing scope-related constraints mainly stimulates innovation by fringe firms, raising their contribution to aggregate growth through stronger entry and competitive pressure, while simultaneously weakening innovation incentives for superstar firms; as a result, the net contribution of incumbents is attenuated and can become negative.

In Section 6, I evaluate three policy instruments targeting the misallocation identified in the counterfactual analysis. The results indicate that a policy combining subsidies for narrow-scope superstar firms with taxes on broad-scope incumbents, complemented by a subsidy to fringe entrants, can generate gains in both consumption and growth. By contrast, policies that primarily target long-run growth tend to involve sizable short-run consumption trade-offs. The analysis also highlights the role of the fringe subsidy in restoring competitive pressure at the entry margin, yielding aggregate effects that are not replicated by incumbent-focused instruments alone.

Related Literature. A growing literature documents that intangible investment increases market concentration and markups (Chiavari and Goraya, 2025; Crouzet and Eberly, 2019; Weiss, 2020). De Ridder (2024) treats software as a firm-specific fixed cost and shows that firms with higher software capability strategically increase their fixed-cost investment to lower their marginal production costs, which creates entry barriers. Aghion *et al.* (2023) distinguish between product and process innovation and they document that declining overhead costs or rising productivity efficiency are the primary forces behind increasing market concentration. In both frameworks, intangible capabilities are treated as a firm's inborn characteristics rather than endogenous investment choices. Cavenaile and Roldan-Blanco (2021) and Cavenaile *et al.* (2025a) show that advertising can substitute for R&D and dampen innovation intensity, a finding further extended by Pearce and Wu (2024), who show that brand value can be transferred in the secondary market rather than only accumulated within the firm. This paper contributes to the literature by establishing that intangible composition is an endogenous determinant of innovation direction and evolves endogenously with firm scope.

The literature has predominantly explained firms' choice of innovation direction through differences in firm size. Akcigit and Kerr (2018) show that smaller firms are more likely to engage in external innovation while larger firms tend toward quality-improving vertical innovation. Berlingieri *et al.* (2025) documents that firms expand their size by adding a series of products simultaneously in the early period, while Garcia-Macia *et al.* (2019) finds that most within-firm innovation arises from incumbents improving existing products rather than entering new lines. I show that intangible composition is an important channel to innovation direction that operates independently of firm size. Firms can choose different innovation paths depending on their allocation of investment across transferable and embedded intangible assets, which firm size alone cannot account for the observed heterogeneity in innovation strategies.

This paper provides a complementary explanation for documented trends, including declining knowledge spillovers (Akcigit and Ates, 2021; Akcigit and Ates, 2023), lower patent quality (Olmstead-Rumsey, 2019), production lock-in (Casal, 2024), and rising entry barriers (Gutiérrez and Philippon, 2019). As the firm's scope expands, investment shifts from transferable toward embedded intangibles, depressing knowledge spillovers and raising entry barriers.

This paper also contributes to the misallocation literature (Hsieh and Klenow, 2009;Restuccia and Rogerson, 2008) by identifying two distinct channels through which intangible heterogeneity generates allocative distortions. The first operates through markup dispersion (Peters, 2020; Edmond *et al.*, 2023), driven by the accumulation of both transferable and embedded intangibles. The second operates through entry barriers: the joint expansion of embedded intangibles and firm scope makes creative destruction asymmetric, generating persistent misallocation that markup-based measures substantially understate.

Finally, the literature on span-of-control has focused primarily on hierarchical organization, knowledge-flow frictions, and managerial ability (Smeets *et al.*, 2019; Bandiera *et al.*, 2014; Garicano, 2000; Bloom and Van Reenen, 2007) and with (Jovanovic, 2025) linking span-of-control constraints to the firm-size distribution at the macro level. I contribute to this literature by disaggregating firm size into vertical and horizontal dimensions of innovation and showing that span-of-control frictions arise only from horizontal expansion. This distinction isolates how scope-driven managerial constraints shape intangible investment, innovation direction, and aggregate growth.

2. Theoretical Model

The theoretical framework builds on the step-by-step vertical innovation framework from Akcigit and Ates (2023) and on the vertical and horizontal innovation frameworks from Peters (2020). I combine both frameworks by introducing two distinct types of intangible investment and span-of-control frictions that arise exclusively from expanding scope.

2.1. Economic Environment

Time is continuous and indexed by t. Household preferences are described by a logarithmic utility function

$$\int_0^\infty e^{-\rho t} \ln(C_t)\, dt, \tag{1}$$

where C_t denotes household consumption and $\rho > 0$ is the time discount rate. The household budget constraint is

$$\dot{A}_t = r_t\, A_t + w_t - C_t, \tag{2}$$

with A_t denotes total household assets, w_t is the wage rate, and r_t is the interest rate. Labor is supplied inelastically, and the price of the consumption good is normalized to 1, where w_t and r_t are expressed in units of the consumption good. Since households own all firms in the economy, total assets equal the aggregate value of superstar and fringe firms across all intermediate-good sectors

$$A_t = \int_0^1 \left(V_{sjt} + V_{fjt}\right) dj,$$

with V_{sjt} and V_{fjt} denoting the values of the superstar and fringe firms.

The final good is produced by aggregating a continuum of intermediate varieties $j \in [0, 1]$ according to

$$\ln(Y_t) = \int_0^1 \ln(y_{jt})\, dj, \tag{3}$$

in a perfectly competitive market. In each intermediate-good sector, a single superstar firm and a continuum of fringe firms compete à la Bertrand to supply the final good producer. A superstar firm s operates across a set of sectors $J_s \subseteq [0,1]$, with $n_s = |J_s| \in \mathbb{Z}_+$ denoting the number of sectors in which it holds the leading technology. Output in each sector is aggregated with a constant elasticity of substitution technology

$$y_{jt} = \left(\Xi(e_{st})\, y_{sjt}^{\varepsilon} + \left(1 - \Xi(e_{st})\right) y_{fjt}^{\varepsilon}\right)^{\frac{1}{\varepsilon}} \tag{4}$$

where $\Xi(e_{st}) \equiv \dfrac{\chi(e_{st})}{1+\chi(e_{st})}$ denotes perceived quality of superstar firms, and $\varepsilon \in (0,1)$. In each sector, the superstar firm produces a differentiated good whose perceived quality is increasing in its brand value. Specifically, $\chi(e_{st}) = (\xi e_{st})^{\beta}$ is an endogenous and concave demand shifter, where e_{st} denotes the embedded intangibles of superstar firm s, $\xi \in (0,1)$ is the share of embedded intangibles associated with brand value, and $\beta \in (0,1)$ governs the curvature of the demand shifter. The term $\Xi(e_{st})$ captures the notion that a higher stock of brand value raises the perceived quality of the superstar's product relative to the fringe. For simplicity, the brand value of fringe firms is normalized to one, and they produce a homogeneous good aggregated as

$$y_{fjt} = \int_0^1 y_{ijt}\, di, \tag{5}$$

where each fringe firm $i \in (0,1)$ takes the market price as given.

The production function of superstar firm s in sector j at time t is

$$y_{sjt} = \underbrace{q_{sjt}}_{\text{Product Quality}} \quad \underbrace{\psi(e_{st}, n_{st})}_{\text{Managerial Productivity}} \quad \underbrace{l_{sjt}}_{\text{Labor Input}}, \tag{6}$$

where q_{sjt} denotes product quality, l_{sjt} is labor input, and $\psi(e_{st}, n_{st})$ captures managerial productivity, defined as

$$\psi(e_{st}, n_{st}) = \frac{\left((1-\xi)\, e_{st}\right)^{\alpha}}{\gamma\, n_{st}^{\alpha}}.$$

The component $(1-\xi)e_{st}$ represents the share of embedded intangibles allocated to organizational capital, which governs managerial efficiency. The variable n_{st} denotes the number of sectors operated by firm s at time t. As firm scope n_{st} increases, managerial productivity per sector declines due to the span-of-control friction: the firm's managerial attention is distributed across a larger number of sectors, reducing the managerial capacity allocated to each individual sector. The parameter α controls the curvature of both organizational capital and the span-of-control constraint, while γ is a scale parameter. Aggregating the managerial productivity of superstar firm s across all active sectors generates total managerial capacity[6]

$$\Lambda \cdot e_{st}^{\alpha} \cdot n_{st}^{1-\alpha}, \quad \text{with} \quad \Lambda \equiv \frac{(1-\xi)^{\alpha}}{\gamma}.$$

When $0 < \alpha < 1$, total managerial capacity increases with both organizational capital and firm scope, yet each exhibits diminishing marginal returns.[7] Total managerial capacity determines the incumbent superstar's ability to defend its market position against challenger entry, a mechanism formalized in the creative destruction section.

Fringe firms produce output using a linear technology

$$y_{fjt} = q_{fjt}\, l_{fjt}, \tag{7}$$

where q_{fjt} denotes product quality and l_{fjt} is labor input. They operate in a single sector, cannot accumulate embedded intangibles, and have managerial quality normalized to one. Their product quality is inherited: when a superstar firm exits a sector, its transferable intangibles become publicly available and fringe firms adopt the previous superstar's quality level.

Superstar firms pursue three distinct innovation strategies. They can invest in transferable intangibles to either improve the quality of an existing product through vertical innovation or expand into a new sector through horizontal innovation. Additionally, they can invest in embedded intangibles to raise their brand value and organizational cap-

[6]Managerial productivity is symmetric across sectors, so total managerial capacity equals per-sector managerial productivity $\psi(e_{st}, n_{st})$ multiplied by the number of active sectors n_{st}.

[7]When $\alpha > 1$, diseconomies of scope emerge: coordination costs outweigh the benefits of expansion, and total managerial capacity decreases with firm scope.

ital across all sectors in which they operate (see Figure 1). The first two strategies require sector-specific investment because product quality improvements embody knowledge specific to each sector's technology. By contrast, embedded intangible investment is firm-specific: an improvement in brand value or organizational capital simultaneously raises the embedded intangible stock of all sectors within the firm's portfolio, reflecting the economies of scope that embedded intangibles generate.

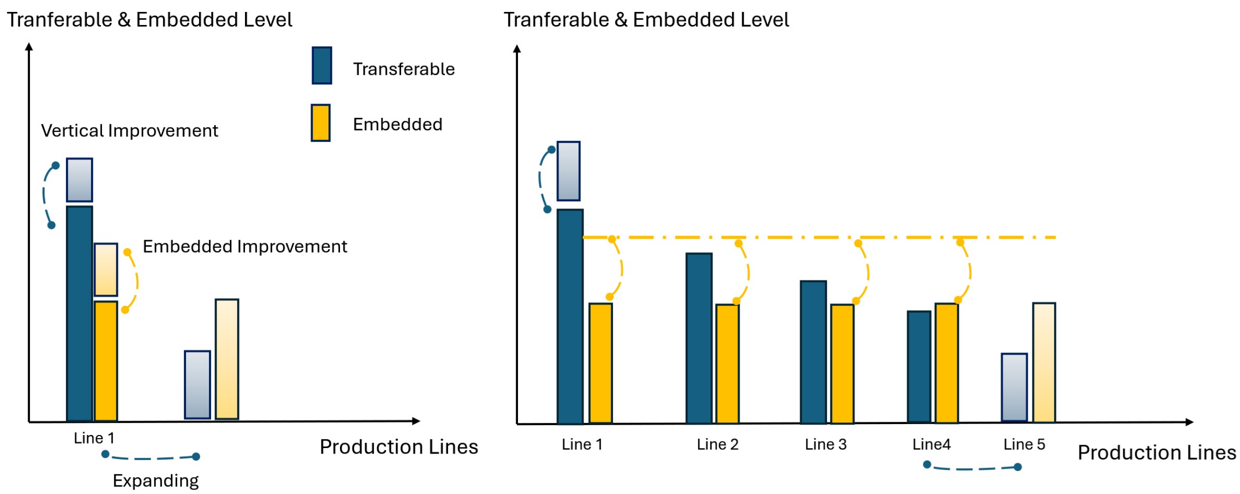

Figure 1. Firm Investment and Innovation Types

The variables I_{sjt}^{Emb}, I_{sjt}^{Ver}, and I_{sjt}^{Hor} denote the investment of superstar firm s in embedded intangibles, vertical innovation in an existing sector, and horizontal innovation into a new sector, respectively. Each unit of investment generates a successful flow rate of innovation: z_{jt}^{Emb} for embedded intangibles, z_{jt}^{Ver} for vertical, and z_{jt}^{Hor} for horizontal innovation investment. Investments are subject to convex costs:

$$I_{sjt}^{\mathrm{Ver}} = \gamma^{\mathrm{Ver}} \left(z_{sjt}^{\mathrm{Ver}}\right)^{\vartheta^{\mathrm{Ver}}} Y_t, \qquad I_{sjt}^{\mathrm{Hor}} = \gamma^{\mathrm{Hor}} \left(z_{sjt}^{\mathrm{Hor}}\right)^{\vartheta^{\mathrm{Hor}}} Y_t, \tag{8}$$

$$\text{and} \qquad I_{sjt}^{\mathrm{Emb}} = \gamma^{\mathrm{Emb}} \left(z_{sjt}^{\mathrm{Emb}}\right)^{\vartheta^{\mathrm{Emb}}} Y_t,$$

where the cost parameters γ^{Ver}, γ^{Hor}, and γ^{Emb} govern the scale of each investment cost function, ϑ^{Ver}, ϑ^{Hor}, and ϑ^{Emb} determine their curvature, and all costs scale with aggregate output Y_t.

Fringe firms invest only in transferable intangibles within their sector, targeting radical innovations that would allow them to displace the incumbent superstar. Their investment cost function takes the form

$$I_{fjt} = \gamma^f \left(z_{fjt}\right)^{\vartheta^f} Y_t, \tag{9}$$

where γ^f is the cost scale parameter and ϑ^f determines the curvature of investment. Taken together, the total investment in transferable intangibles in sector j at time t combines vertical and horizontal investments by the incumbent superstar and investment by fringe firms:

$$I_{jt}^{\mathrm{T}} = I_{sjt}^{\mathrm{Ver}} + I_{sjt}^{\mathrm{Hor}} + I_{fjt}. \tag{10}$$

A successful quality improvement — whether vertical within an existing sector or horizontal into a new sector — improves product quality by a factor of $\lambda > 1$, while a successful embedded intangible investment raises brand value and organizational capital by a factor of $\theta > 1$ across all active sectors simultaneously. Product quality and embedded intangibles therefore evolve as

$$q_{sjt} = \lambda^{m_{sjt}}\, q_{sj0}, \quad \text{and} \quad e_{st} = \theta^{k_{st}}\, e_{s0}, \tag{11}$$

where m_{sjt} denotes the cumulative number of product-quality innovations by firm s in sector j up to time t, and k_{st} denotes the cumulative number of embedded intangible innovations by firm s up to time t, with initial levels normalized to $q_{sj0} = 1$ and $e_{s0} = 1$.

The quality gap between the superstar and the fringe in sector j at time t is therefore

$$\frac{q_{sjt}}{q_{fjt}} = \lambda^{m_{sjt} - m_{fjt}} = \lambda^{m_{jt}}, \tag{12}$$

where $m_{jt} \equiv m_{sjt} - m_{fjt}$ denotes the transferable intangible gap. Since the embedded intangible stock of fringe firms is normalized to one, the embedded intangible gap simplifies to the superstar's own stock:

$$\frac{e_{st}}{e_{ft}} = \frac{\theta^{k_{st}}}{1} = \theta^{k_{st}}. \tag{13}$$

Assumption (Managerial Reallocation Capability). *When a superstar firm — whether incumbent or challenger — faces competition, it can temporarily concentrate its entire managerial capacity on the contested sector. This reallocation is costless in the short run and determines the outcome of the price competition stage.*

Creative destruction arises either from a challenger superstar firm expanding its scope or from a fringe firm pursuing radical innovation. When a superstar firm successfully pursues horizontal innovation, it enters a randomly selected sector j', improving product quality by a factor of λ such that $q_{sjt} = \lambda\, q_{s'jt}$, and displaces the incumbent if its effective unit cost — defined as the wage relative to total managerial capacity — falls below that of the incumbent:

$$\frac{w_t}{\Lambda\, q_{sjt}\, e_{st}^{\alpha}\, n_{st}^{1-\alpha}} < \frac{w_t}{\Lambda\, q_{s'jt}\, e_{s't}^{\alpha}\, n_{s't}^{1-\alpha}}.$$

The probability that the entrant successfully displaces the incumbent is

$$\mathbb{P}_{s>s'} \;\equiv\; \sum_{m_{s'}=1}^{\bar{m}} \sum_{k_{s'}=1}^{\bar{k}} \sum_{n_{s'}=1}^{\bar{n}} \mathbb{I}\big\{\lambda\, (\theta^{k_s})^{\alpha}\, n_{st}^{1-\alpha} > (\theta^{k_{s'}})^{\alpha}\, n_{s't}^{1-\alpha}\big\}\, \mu_t(m_{s'}, k_{s'}, n_{s'}),$$

where s denotes the entrant and s' the incumbent, with the left-hand side reflecting the entrant's total managerial capacity after the quality improvement and the right-hand side the incumbent's. Here $\mu_t(m_{s'}, k_{s'}, n_{s'}) \in [0,1]$ denotes the mass of superstar firms in state $(m_{s'}, k_{s'}, n_{s'})$ at time t. The state space is finite and discrete, given by $\mathcal{M} \times \mathcal{K} \times \mathcal{N}$, with $\mathcal{M} = \{1, \ldots, \bar{m}\}$, $\mathcal{K} = \{1, \ldots, \bar{k}\}$, and $\mathcal{N} = \{1, \ldots, \bar{n}\}$ with satisfies

$$\sum_{m_{s'},\, k_{s'},\, n_{s'}} \mu_t(m_{s'}, k_{s'}, n_{s'}) = 1. \tag{14}$$

The probability of losing the specific sector j' is given by

$$\mathbb{P}_{c>s'} = \sum_{m_c,\, k_c,\, n_c} \mathbb{I}\big\{\lambda\, (\theta^{k_c})^{\alpha}\, n_{ct}^{1-\alpha} > (\theta^{k_s})^{\alpha}\, n_{st}^{1-\alpha}\big\}\; \mu_t(m_c, k_c, n_c).$$

Fringe firms cannot accumulate embedded intangibles and therefore cannot compete with the incumbent on managerial capacity. Their only way to be a superstar is through radical innovation, which requires jumping multiple steps.[8] on the quality ladder by a factor $\lambda^{\bar{m}} > \lambda$, such that $q_{sjt} = \lambda^{\bar{m}} q_{s'jt}$[9] Since fringe firms are symmetric and invest identically within each sector, a fringe firm successfully displaces the incumbent in sector j if and only if

$$\mathbb{I}_{f>s'} \equiv \mathbb{I}\left\{\lambda^{\bar{m}} > (\theta^{k_{s'}})^{\alpha} n_{s'}^{1-\alpha}\right\},$$

where the right-hand side depends solely on the incumbent's state $(k_{s'}, n_{s'})$. Entry conditions through creative destruction show that the barrier is increasing in both the incumbent's embedded intangible stock $\theta^{k_{s'}}$ and its scope $n_{s'}$, regardless of whether the challenger is a fringe firm or superstar. Moreover, two embedded intangibles play asymmetric roles in the entry condition. Since only organizational capital enters total managerial capacity, eliminating it reduces entry barriers. By contrast, eliminating brand value raises entry barriers, as any increase in the embedded intangible stock then translates one-to-one into a stronger competitive advantage against challengers.

2.2. Equilibrium

The equilibrium has a static and a dynamic component. The analysis begins with the static equilibrium, determining prices and allocations for a given set of states. I then define the Markov Perfect Equilibrium for the dynamic game, outlining the value functions, optimal policy functions, and the evolution of the aggregate state distribution.

A household maximizes utility subject to the budget constraint, yielding the Euler equation:

$$\frac{\dot{C}_t}{C_t} = r_t - \rho. \tag{15}$$

Along the balanced growth path, consumption and output grow at the same rate $g = r - \rho$, and the transversality condition holds. The final-good producer's demand for intermedi-

[8]Fringe firms in the model correspond to small firms in the data. A vast literature documents that small and young firms grow faster than their larger counterparts (see Evans, 1987, Haltiwanger *et al.*, 2013).

[9]Following Akcigit *et al.* (2026), drastic innovation by fringe firms can alternatively be modeled using a probability distribution $\mathbb{F}(\cdot)$ over the innovation jumps, rather than a fixed step of size $\bar{m}$.

ate goods in sector j satisfies

$$p_{jt} \;=\; \frac{Y_t}{y_{jt}}. \tag{16}$$

This implies the following demand functions for the superstar and fringe firms:

$$y_{sjt} = p_{jt}^{\frac{\varepsilon}{1-\varepsilon}} p_{sjt}^{\frac{1}{\varepsilon-1}} Y_t \big(\Xi(e_s)\big)^{\frac{1}{1-\varepsilon}} \quad \text{and} \quad y_{fjt} = p_{jt}^{\frac{\varepsilon}{1-\varepsilon}} p_{fjt}^{\frac{1}{\varepsilon-1}} Y_t \big(1-\Xi(e_s)\big)^{\frac{1}{1-\varepsilon}}, \tag{17}$$

where p_{sjt} and p_{fjt} are the prices charged by the superstar and fringe firms, respectively. The corresponding ideal price index for sector j is given by

$$p_{jt} = \left(\big(\Xi(e_s)\big)^{\frac{-1}{\varepsilon-1}} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} + \big(1-\Xi(e_s)\big)^{\frac{-1}{\varepsilon-1}} p_{fjt}^{\frac{\varepsilon}{\varepsilon-1}} \right)^{\frac{\varepsilon-1}{\varepsilon}}. \tag{18}$$

The Cobb-Douglas aggregator implies equal expenditure shares across all sectors. The market share of superstar firm s in sector j at time t is

$$\frac{p_{sjt}y_{sjt}}{p_{jt}y_{jt}} = \frac{p_{sjt}y_{sjt}}{Y_t} = p_{jt}^{\frac{\varepsilon}{1-\varepsilon}} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} \, \Xi(e_s)^{\frac{1}{1-\varepsilon}} \;\equiv\; \phi_{sjt}, \tag{19}$$

and since market shares sum to one, the fringe firms' share is $1-\phi_{sjt}$. The equilibrium price of the superstar firm under Bertrand competition is[10]

$$p_{sjt} = \frac{1-\varepsilon\,\phi_{sjt}}{(1-\phi_{sjt})\,\varepsilon} \; \underbrace{\frac{w_t\, n_{st}^{\alpha}}{\Lambda\, q_{sjt}\, e_{st}^{\alpha}}}_{MC_{sjt}}, \tag{20}$$

where MC_{sjt} denotes the marginal cost of the superstar firm.[11] The equilibrium price increases with product quality and the embedded intangible stock, and decreases with the number of sectors operated. The price ratio of fringe to superstar firms is

$$\frac{p_{fjt}}{p_{sjt}} = \frac{(1-\phi_{sjt})\varepsilon}{1-\varepsilon\phi_{sjt}} \cdot \lambda^{m_j} \left(\frac{(1-\xi)e_{st}}{n_{st}} \right)^{\alpha}. \tag{21}$$

[10]See Appendix B.3 for the full derivations. For the à la Cournot competition, see Appendix B.4.

[11]Although fringe firms lack independent pricing power, their presence creates a competitive constraint that limits the superstar firm's pricing power and prevents it from extracting monopolistic rents.

Substituting the ideal price index from equation (18) into the market share definition in equation (19), the superstar's market share can be expressed in terms of relative prices as

$$\phi_{sjt} = \frac{1}{1+\left(\frac{1}{(\chi(e_{st}))^{\frac{1}{1-\varepsilon}}}\left(\frac{p_{fjt}}{p_{sjt}}\right)^{\frac{\varepsilon}{\varepsilon-1}}\right)}, \tag{22}$$

which depends on the quality gap m_j, the embedded intangible stock e_{st}, and the number of sectors operated n_{st}.

The operational profit and markup of the superstar firm are proportional to its market share:

$$\pi_{sjt} = \frac{(1-\varepsilon)\,\phi_{sjt}}{1-\varepsilon\phi_{sjt}}\,Y_t \qquad \text{and} \qquad \sigma_{sjt} = \frac{1-\varepsilon\phi_{sjt}}{(1-\phi_{sjt})\,\varepsilon}. \tag{23}$$

Both profit and markup increase with market share and therefore decrease with firm scope, reflecting the diminishing managerial productivity that arises from the span-of-control friction. The optimal labor inputs for superstar and fringe firms are

$$l_{sjt} = \frac{\phi_{sjt}}{\sigma_{sjt}}\,\omega_t^{-1} \quad \text{and} \quad l_{fjt} = (1-\phi_{sjt})\,\omega_t^{-1}, \tag{24}$$

where $\omega_t = w_t/Y_t$ denotes the wage share of the economy. The static equilibrium provides only an implicit solution. Nonetheless, the model yields tractable dynamics because the equilibrium outcomes for both superstar and fringe firms depend solely on their market share, which is determined by the quality gap, the embedded intangible stock, and the number of sectors in which it operates.

The value function of superstar firm s, which depends on the quality gap vector $\mathbf{m} = \{m_j\}_{j=1}^{n}$, the embedded intangible level k, and the number of sectors operated n.[12] The

[12]To simplify notation, the subscript s is omitted whenever it is clear from context.

firm chooses innovation flow rates z_j^{Ver}, z_j^{Hor}, and z_j^{Emb} to maximize

$$
\begin{aligned}
r_t V_t(\mathbf{m},k,n) - \dot{V}_t(\mathbf{m},k,n) = \max_{z_{jt}^{\text{Ver}}, z_{jt}^{\text{Hor}}, z_{jt}^{\text{Emb}}} \sum_{j=1}^{n} \Bigg(& \frac{(1-\varepsilon)\,\phi_t(m_j,k,n)}{1-\varepsilon\,\phi_t(m_j,k,n)} Y_t \\
& + \underbrace{\mathbb{P}_{s\geq s'}\, z_{jt}^{\text{Hor}}\Big[V_{jt}\big((m_j,1),k,n+1\big) - V_{jt}(m_j,k,n)\Big]}_{\text{Horizontal Innovation}} + \underbrace{z_{jt}^{\text{Ver}}\Big[V_{jt}(m_j+1,k,n) - V_{jt}(m_j,k,n)\Big]}_{\text{Vertical Innovation}} \\
& + \underbrace{z_{jt}^{\text{Emb}}\Big[V_{jt}(\mathbf{m},k+1,n) - V_{jt}(\mathbf{m},k,n)\Big]}_{\text{Embedded Innovation}} + \underbrace{Z_{c>s\,t}^{\text{Hor}}\Big[-V_{jt}(m_j,k,n)\Big]}_{\text{Challenger Superstar Entry}} + \underbrace{\mathbb{I}_{f>s}\,Z_{jt}^{f}\Big[-V_{jt}(m_j,k,n)\Big]}_{\text{Fringe Entry}} \\
& - \gamma^{\text{Ver}}\left(z_{jt}^{\text{Ver}}\right)^{\vartheta^{\text{Ver}}} Y_t - \gamma^{\text{Emb}}\left(z_{jt}^{\text{Emb}}\right)^{\vartheta^{\text{Emb}}} Y_t - \gamma^{\text{Hor}}\left(z_{jt}^{\text{Hor}}\right)^{\vartheta^{\text{Hor}}} Y_t \Bigg).
\end{aligned}
\tag{25}
$$

The left-hand side represents the flow return on the firm's value. The first term on the right-hand side is the operational profit of the superstar firm in sector j. The second and third terms capture the value gains from horizontal and vertical innovation, respectively: a successful horizontal innovation expands the firm's scope from n to $n+1$, while a successful vertical innovation advances the quality gap in sector j by one rung. The fourth term captures the superstar gain from a successful embedded intangible investment, which raises the embedded gap k by one rung across all active sectors simultaneously. The fifth and sixth terms capture the value loss upon displacement by a challenger superstar or a fringe firm. The final three terms are the convex investment costs defined in equation (8). The terms $Z_{c>s\,t}^{\text{Hor}}$ and Z_{jt}^{f} denote the aggregate innovation rates of challenger superstar firms and fringe firms, respectively. The aggregate horizontal innovation rate of challenger superstars that successfully displace the incumbent is

$$
Z_{c>s\,t}^{\text{Hor}} = \sum_{m_c,k_c,n_c} \mathbb{I}\big\{\lambda\,(\theta^{k_c})^{\alpha}\, n_{ct}^{1-\alpha} > (\theta^{k_s})^{\alpha}\, n_{st}^{1-\alpha}\big\}\, z_t^{\text{Hor}}(m_c,k_c,n_c)\,\mu_t(m_c,k_c,n_c),
$$

and the aggregate fringe firm innovation rate is $Z_{jt}^{f} = \int_0^1 z_{ijt}\,di$.

Along the balanced growth path, aggregate output Y, consumption C, and the value function $V(\mathbf{m},k,n)$ all grow at the constant rate g. Dividing the value function by aggregate output and defining $v(\mathbf{m},k,n) \equiv V(\mathbf{m},k,n)/Y$ yields the normalized superstar value function, which in the stationary equilibrium satisfies

$$
\begin{aligned}
\rho\, v(\mathbf{m},k,n) = \max_{z_j^{\text{Ver}}, z_j^{\text{Hor}}, z_j^{\text{Emb}}} \sum_{j=1}^{n} \Bigg(& \frac{(1-\varepsilon)\,\phi(m_j,k,n)}{1-\varepsilon\,\phi(m_j,k,n)} \\
& + \underbrace{\mathbb{P}_{s\geq s'}\, z_j^{\text{Hor}} \Big[v\big((m_j,1),k,n+1\big) - v(m_j,k,n) \Big]}_{\text{Horizontal Innovation}} + \underbrace{z_j^{\text{Ver}} \Big[v(m_j+1,k,n) - v(m_j,k,n) \Big]}_{\text{Vertical Innovation}} \\
& + \underbrace{z_j^{\text{Emb}} \Big[v(\mathbf{m},k+1,n) - v(\mathbf{m},k,n) \Big]}_{\text{Embedded Innovation}} + \underbrace{Z_{c>s}^{\text{Hor}} \Big[-v(m_j,k,n) \Big]}_{\text{Challenger Superstar Entry}} + \underbrace{\mathbb{I}_{f>s}\, Z_j^{f} \Big[-v(m_j,k,n) \Big]}_{\text{Fringe Entry}} \\
& - \gamma^{\text{Ver}} \left(z_j^{\text{Ver}} \right)^{\vartheta^{\text{Ver}}} - \gamma^{\text{Emb}} \left(z_j^{\text{Emb}} \right)^{\vartheta^{\text{Emb}}} - \gamma^{\text{Hor}} \left(z_j^{\text{Hor}} \right)^{\vartheta^{\text{Hor}}} \Bigg).
\end{aligned}
\tag{26}
$$

Since fringe firms produce a homogeneous good, they earn zero profit in equilibrium and derive value solely from the possibility of displacing the incumbent through radical innovation. The fringe firm chooses innovation intensity z_f to maximize this value, trading off investment costs against the gain from becoming the new superstar. The normalized fringe firm value function $v_f(m,k,n) \equiv V_{ft}(m,k,n)/Y$ in the stationary equilibrium satisfies

$$
\begin{aligned}
\rho\, v_f(m,k,n) = \max_{z_f} \Bigg\{ & \underbrace{\mathbb{I}_{f>s'}\, z_f \Big[v(\bar{m},1,1) - v_f(m,k,n) \Big]}_{\text{Radical Innovation}} - \gamma^f (z_f)^{\vartheta^f} \\
& + \underbrace{\mathbb{P}_{s\geq s'}\, z_j^{\text{Hor}} \Big[v_f\big((m,1),k,n+1\big) - v_f(m,k,n) \Big]}_{\text{Incumbent Horizontal Innovation}} + \underbrace{z_j^{\text{Ver}} \Big[v_f(m+1,k,n) - v_f(m,k,n) \Big]}_{\text{Incumbent Vertical Innovation}} \\
& + \underbrace{z_j^{\text{Emb}} \Big[v_f(m,k+1,n) - v_f(m,k,n) \Big]}_{\text{Incumbent Embedded Innovation}} + \underbrace{\mathbb{E}_\mu \big[\mathbb{I}_{c>s} \cdot z^{\text{Hor}} \cdot \Delta v_f \big]}_{\text{Challenger Superstar Entry}} + \underbrace{\mathbb{I}_{f>s}\, Z_j^{f} \Big[v_f(\bar{m},1,1) - v_f(m,k,n) \Big]}_{\text{Fringe Entry}} \Bigg\},
\end{aligned}
\tag{27}
$$

where the challenger superstar entry term is

$$
\mathbb{E}_\mu \big[\mathbb{I}_{c>s} \cdot z^{\text{Hor}} \cdot \Delta v_f \big] \equiv \sum_{m_c,k_c,n_c} \mathbb{I}\big\{ \lambda(\theta^{k_c})^\alpha n_c^{1-\alpha} > (\theta^{k_s})^\alpha n_s^{1-\alpha} \big\}\, z^{\text{Hor}}(m_c,k_c,n_c)\, \mu(m_c,k_c,n_c)\, \Delta v_f .
$$

with $\Delta v_f \equiv v_f(1,k_s,n_s) - v_f(m,k,n)$ denoting the value change for the fringe firm upon facing a new incumbent. The first term captures the value gain from radical innovation: the indicator $\mathbb{I}_{f>s'}$ equals one when the fringe firm's innovation is sufficient to overcome the incumbent's total managerial capacity, upon which it enters as a new superstar with state $(\bar{m}, 1, 1)$. The remaining terms account for changes in the competitive environment arising from the incumbent's own innovations, challenger superstar entry, and successful radical innovation by another fringe firm, each of which alters the state the fringe firm faces.

The first-order conditions of the superstar and fringe value functions in equations (26) and (27) yield the following optimal innovation rates along the balanced growth path:

$$z_j^{\text{Hor}} = \left(\frac{\mathbb{P}_{s \geq s'} \left[v_j\big((m_j, 1), k, n+1\big) - v_j(m_j, k, n) \right]}{\gamma^{\text{Hor}} \cdot \vartheta^{\text{Hor}}} \right)^{\frac{1}{\vartheta^{\text{Hor}} - 1}} \tag{28}$$

$$z_j^{\text{Ver}} = \left(\frac{v_j(m_j + 1, k, n) - v_j(m_j, k, n)}{\gamma^{\text{Ver}} \cdot \vartheta^{\text{Ver}}} \right)^{\frac{1}{\vartheta^{\text{Ver}} - 1}} \tag{29}$$

$$z_j^{\text{Emb}} = \left(\frac{v(\mathbf{m}, k+1, n) - v(\mathbf{m}, k, n)}{\gamma^{\text{Emb}} \cdot \vartheta^{\text{Emb}}} \right)^{\frac{1}{\vartheta^{\text{Emb}} - 1}} \tag{30}$$

$$z_f = \left(\frac{\mathbb{I}_{f>s} \left[v(\bar{m}, 1, 1) - v_f(m, k, n) \right]}{\gamma^f \cdot \vartheta^f} \right)^{\frac{1}{\vartheta^f - 1}} \tag{31}$$

Each optimal innovation rate is increasing in the value gain from the corresponding innovation and decreasing in its cost scale and curvature parameters.

The law of motion of the distribution $\mu_t(m, k, n)$ is driven by net flows into and out of state (m, k, n). Inflows arise from firms that transition into (m, k, n) following a successful innovation along one or more dimensions; outflows arise from firms that leave (m, k, n) as a result of their own innovation or displacement through creative destruction. The first three terms capture inflows from states one step behind in a single dimension; the next three capture inflows from states one step behind in two dimensions; and the subsequent term captures inflows from states one step behind in all three dimensions simultaneously. The final terms capture outflows due to the incumbent's own innovation, displacement by a fringe firm, and displacement by a challenger superstar.[13]

$$
\begin{aligned}
\dot{\mu}_t(m,k,n) = \; & z_t^{\text{Ver}}(m-1,k,n)\,\mu_t(m-1,k,n) + z_t^{\text{Emb}}(m,k-1,n)\,\mu_t(m,k-1,n) \\
& + \mathbb{P}_{s\geq s'}\, z_t^{\text{Hor}}(m,k,n-1)\,\mu_t(m,k,n-1) \\
& + z_t^{\text{Ver}}(m-1,k-1,n)\, z_t^{\text{Emb}}(m-1,k-1,n)\,\mu_t(m-1,k-1,n) \\
& + z_t^{\text{Ver}}(m-1,k,n-1)\,\mathbb{P}_{s\geq s'}\, z_t^{\text{Hor}}(m-1,k,n-1)\,\mu_t(m-1,k,n-1) \\
& + z_t^{\text{Emb}}(m,k-1,n-1)\,\mathbb{P}_{s\geq s'}\, z_t^{\text{Hor}}(m,k-1,n-1)\,\mu_t(m,k-1,n-1) \\
& + z_t^{\text{Ver}}(m-1,k-1,n-1)\, z_t^{\text{Emb}}(m-1,k-1,n-1) \\
& \times\; \mathbb{P}_{s\geq s'}\, z_t^{\text{Hor}}(m-1,k-1,n-1)\,\mu_t(m-1,k-1,n-1) \\
& - z_t^{\text{Ver}}(m,k,n)\,\mu_t(m,k,n) - z_t^{\text{Emb}}(m,k,n)\,\mu_t(m,k,n) \\
& - \mathbb{P}_{s\geq s'}\, z_t^{\text{Hor}}(m,k,n)\,\mu_t(m,k,n) - \mathbb{I}_{f>s}\, Z_t^{f}(m,k,n)\,\mu_t(m,k,n) - Z^{\text{Hor}}_{c>s\,t}\,\mu_t(m,k,n)
\end{aligned}
\tag{32}
$$

The labor market clears according to

$$1 = \int_0^1 \left(l_{sjt} + l_{fjt}\right) dj. \tag{33}$$

Using equations (24) and (33), the normalized wage satisfies

$$\omega_t = \sum_{m=1}^{\bar{m}} \sum_{k=1}^{\bar{k}} \sum_{n=1}^{\bar{n}} \left(\tfrac{\phi_t(m,k,n)}{\sigma_t(m,k,n)} + 1 - \phi_t(m,k,n) \right) \mu_t(m,k,n). \tag{34}$$

Combining the production functions in equations (3), (6), and (7) with the labor demand equations (24) yields aggregate output

$$Y_t = Q_t\,\omega_t^{-1}\, \exp\left(\int_0^1 \ln \left[\Xi\left(e_{st}\right) \left(\frac{((1-\xi)e_{st})^{\alpha}}{\gamma n_{st}^{\alpha_s}} \frac{\phi_{sjt}}{\sigma_{sjt}} \right)^{\varepsilon} + (1-\Xi(e_{st})) \left(\lambda^{-m_{jt}} (1-\phi_{sjt}) \right)^{\varepsilon} \right]^{\frac{1}{\varepsilon}} dj \right), \tag{35}$$

where $Q_t = \exp\Big(\int_0^1 \ln q_{sjt}\, dj \Big)$ denotes the aggregate quality. Along the balanced growth

[13] At the boundaries of the state space, the relevant outflow terms are set to zero. When any dimension attains its upper bound $(\bar{m}, \bar{k}, \bar{n})$, outflows from further innovation in that dimension are zero.

path, the economy grows at the rate[14]

$$g \;=\; \ln\lambda \sum_{m=1}^{\bar{m}}\sum_{k=1}^{\bar{k}}\sum_{n=1}^{\bar{n}} \Big(z^{\mathrm{Ver}}(m,k,n) + \mathbb{P}_{s\geq s'}\, z^{\mathrm{Hor}}(m,k,n) + Z^{f}(m,k,n)\Big)\,\mu(m,k,n). \quad (36)$$

Finally, the resource constraint requires that aggregate output is allocated between consumption and investment across all sectors:

$$Y_t = C_t + \int_0^1 \left(I_{jt}^{\mathrm{Ver}} + I_{jt}^{\mathrm{Hor}} + I_{jt}^{\mathrm{Emb}}\right) dj + \int_0^1 I_{jt}^{f}\, dj, \quad \text{with} \quad I_{jt}^{f} = \int_{F_j} I_{ijt}\, di. \quad (37)$$

Definition (Markov Perfect Equilibrium). *A Markov Perfect Equilibrium consists of an allocation* $\{C_t, Y_t, y_{sjt}, y_{fjt}\}$*, prices* $\{r_t, w_t, p_{sjt}, p_{fjt}\}$*, innovation policies* $\{z^{Ver}, z^{Hor}, z^{Emb}, z_f\}$*, labor allocations* $\{l_{sjt}, l_{fjt}\}$*, and a distribution* $\mu_t(m,k,n)$ *such that:* (i) *the final good producer maximizes profit given prices;* (ii) *the superstar firm maximizes its value function given state* $(\boldsymbol{m}, k, n)$*, choosing innovation rates as in equations* (28)–(30)*;* (iii) *the fringe firm chooses its optimal innovation rate as in equation* (31)*;* (iv) *the labor market clears;* (v) *the distribution* $\mu_t(m,k,n)$ *satisfies equation* (14) *and evolves according to equation* (32)*; and* (vi) *aggregate consumption and output grow at the common rate* g *given by equation* (36)*, with the resource constraint* (37) *satisfied.*

3. Dataset and Empirical Facts

3.1. Data Description

The empirical strategy centers on documenting how markups, productivity, intangible investment composition, and competitive threats vary across the scope distribution. The main dataset is constructed by combining two Compustat sources — Compustat Fundamentals and the Compustat Segment dataset — supplemented by the product market fluidity metric from Hoberg *et al.* (2014) and the patent-value dataset from Kogan *et al.* (2017).

[14] See Appendix B.5 for details.

Compustat Fundamentals provides comprehensive firm-level financial information for publicly listed companies in North America[15], with extensive longitudinal coverage of balance sheet items, income statement components, cash flow data, and key financial ratios. Measuring firm scope additionally requires information on the industries in which a firm operates. Compustat provides two segment datasets: (i) the Historical Segment dataset[16], which offers long-run coverage of firms' segment activities but limited segment-level detail, and (ii) the Compustat Segment dataset, introduced in 2016, which provides richer segment-level information over a shorter time horizon. In both datasets, a firm's segments are defined as the two-digit industries in which it operates.[17] To balance coverage and detail, I combine the two sources, prioritizing the Compustat Segment dataset when available and relying on the Historical Segment dataset otherwise.[18] I define firm scope as the number of two-digit industries in which a firm operates in a given year, and merge this classification with Compustat Fundamentals to construct the firm-level measures of markups, productivity, and intangible investment composition that form the basis of the empirical analysis.

To capture the competitive threat firms face, I use the product-market fluidity metric from Hoberg *et al.* (2014), which measures how rapidly rivals in similar markets change their product or service offerings. The metric is constructed using natural language processing on product descriptions from firms' annual 10-K filings with the U.S. Securities and Exchange Commission, tracking year-over-year changes in how companies describe

[15]Compustat North America includes foreign firms cross-listed in the U.S. via American Depositary Receipts (ADRs), such as Toyota and Unilever. These firms are subject to SEC regulation, comply with U.S. reporting requirements, and actively compete in U.S. product markets, making them economically relevant for the study of firm dynamics. As their exclusion reduces the sample size by approximately 15% without altering the main conclusions, they are retained throughout the analysis. Table A2 provides a comparison of summary statistics.

[16]Under Regulation SFAS No. 131 — codified as ASC 280 after 2009 — U.S. public firms are required to disclose the identity of any customer accounting for more than 10% of total revenue, along with the nature of products or services provided. These mandated disclosures constitute the foundation of the Historical Segment dataset.

[17]For a comparative illustration of firm segment classification, see Table A1.

[18]From the Historical Segment dataset, geographic and operational segments are excluded; only business segments are retained. From the Compustat Segment dataset, only non-missing entries from the Product–Service (`PD-SRVC`) category are included.

their business activities. A high fluidity score indicates that rivals are rapidly reconfiguring their market positions, intensifying the competitive threat the firm faces.

To trace how firms respond to innovation, I use the patent-value dataset from Kogan *et al.* (2017), which provides patent identifiers, filing and grant dates, firm identifiers, forward citations, and a measure of patent value. The patent value measure is constructed from stock market reactions within a narrow event window around patent grant dates, capturing the surprise component of the market's response to each grant. This short-window design isolates the incremental information content of the grant itself, abstracting from anticipation effects and concurrent firm-level news.

3.2. Measurement

I estimate firm-level total factor productivity following Olley and Pakes (1996), who resolve the simultaneity bias inherent in OLS-based production function estimation by exploiting the monotonicity of investment in unobserved productivity to control for the endogeneity of input choices.[19] To estimate firm-level markups, I follow De Loecker *et al.* (2020) and define markups as the ratio of the output elasticity of the variable input to its expenditure share in total sales, where the output elasticity is recovered from the first-stage production function estimation.

Following Peters and Taylor (2017), I measure transferable intangible investment as total R&D expenditure and embedded intangible investment as 30% of Selling, General, and Administrative (SG&A) expenses net of R&D. SG&A encompasses a broad range of expenditures including advertising, employee compensation, and general operational costs. Once R&D is netted out, the remaining SG&A primarily reflects advertising spending and employee-related costs, most of which are routine operating expenses consumed in the current period. However, a portion accumulates within the firm as lasting intangible capital: advertising builds brand recognition and employee compensation develops organizational knowledge and managerial talent. The 30% fraction captures this accumu-

[19] For methodological details, see Appendix A.2.1. Table A3 reports output elasticity estimates under alternative production function estimators, following Ackerberg *et al.* (2015) and Gandhi *et al.* (2020).

lating component, with the remainder treated as routine operating costs that generate no lasting intangible value.[20]

3.3. Empirical Facts

To document how firm-level outcomes respond to innovation, I estimate local projections following Jordà (2005), using the patent-value measure of Kogan *et al.* (2017) as an innovation shock. This measure is well-suited for the present analysis: its market-based construction isolates the unexpected component of each grant, grant timing is largely predetermined relative to the firm's strategic decisions, and it assigns a dollar value to each patent, capturing the monetary magnitude of each innovation rather than treating all grants as qualitatively equivalent.[21] The local projection specification takes the form

$$\Delta_h Y_{ijt} = \alpha_h + \beta_{\text{Multi}}\, V_{it} D_{it} + \beta_{\text{Single}}\, V_{it}(1 - D_{it}) + \mathbf{\Gamma}_h^{\top} \mathbf{X}_{i,t-1} + \delta_h Y_{i,t-1} + \theta_j + \lambda_t + u_{ijt}, \quad (38)$$

where $\Delta_h Y_{ijt} \equiv Y_{i,t+h} - Y_{it}$ is the h-period change in outcome Y for firm i in industry j, $V_{it} = \sum_{p \in \mathcal{P}_{it}} \text{PatentValue}_{p,t}$ is the firm-year sum of innovation shocks, and $D_{it} = \mathbf{1}\{i \text{ is multi-sector at } t\}$ indicates multi-sector status. The vector $\mathbf{X}_{i,t-1}$ collects lagged controls, $Y_{i,t-1}$ absorbs pre-shock outcome levels, and θ_j and λ_t denote industry and year fixed effects. The coefficients β_{Single} and β_{Multi} trace the impulse responses of single- and multi-sector firms, respectively, to an equivalent innovation shock. Scope status is defined using each firm's pre-shock classification and held fixed throughout the estimation window, ensuring that firms that subsequently change their reported segments do not mechanically alter the interaction term.

[20] The two components of embedded intangibles — brand value and organizational capital — could be identified separately, as advertising expenditure is reported as a distinct line item (XAD) in Compustat. However, XAD is sparsely reported in Compustat, and conditioning on non-missing values would disproportionately retain firms with large advertising budgets, introducing a systematic selection bias toward advertising-intensive firms.

[21] Segment changes are infrequent in the data, suggesting that the innovation shocks identified here predominantly reflect vertical improvements within existing sectors rather than scope-expanding horizontal innovations.

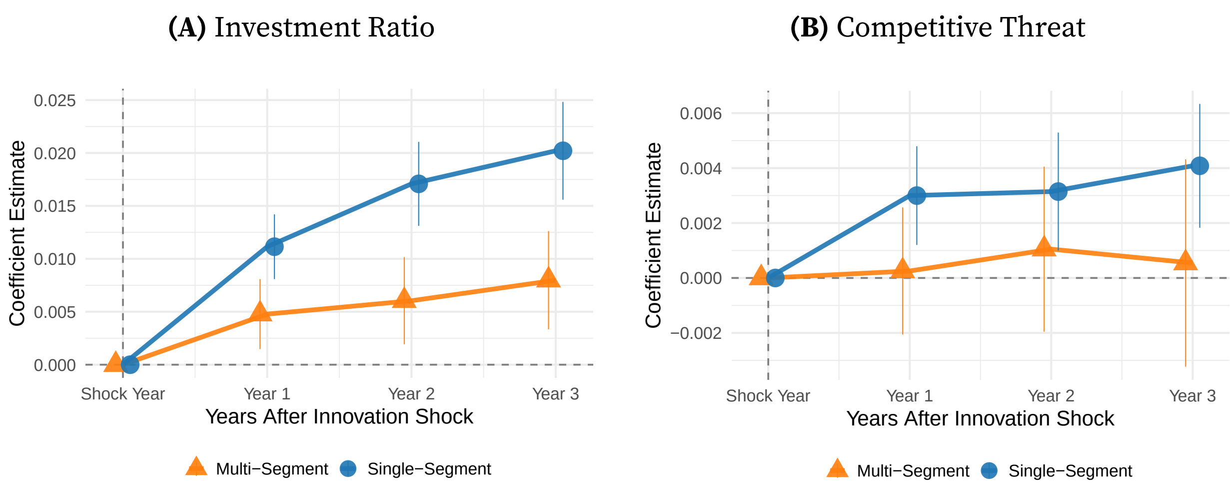

Figure 2. Investment Ratio and Competitive Threat Responses to Innovation Shocks

Note: Note: The innovation shock is measured as $\log(1 + V_{it})$, preserving firms with zero patent grants in a given year. Outcome variables and controls are measured in $\log(\cdot)$. The sample excludes utilities and finance sectors, as well as firms with missing or non-positive R&D, SG&A, employment, and sales. All variables cover the period 1990–2019. Vertical bars denote 95% confidence intervals based on standard errors clustered at the industry-year level.

Innovation shocks produce qualitatively similar responses across firm types, yet the degree of response differs substantially and consistently across every margin examined. Single-sector firms reallocate investment sharply toward transferable intangibles following a shock, with the response emerging immediately after the shock and widening monotonically over the subsequent three years (Figure 2A). Multi-sector firms exhibit a substantially weaker reallocation, suggesting that operating across multiple sectors dilutes the incentive to concentrate innovative investment in a single domain.

The competitive consequences of innovation reinforce this pattern. Single-sector innovators attract significant rival entry following a shock, indicating that their innovations diffuse outward and reshape the competitive landscape, while multi-sector firms experience no discernible increase in competitive threat (Figure 2B). This near-zero response admits two complementary interpretations: multi-sector firms may erect stronger entry barriers through the breadth of their market presence, or their innovations may rest on cross-sector complementarities that narrower rivals cannot easily replicate. Either mechanism implies that multi-sector firms appropriate a larger share of innovation gains internally, insulating themselves from the competitive spillovers that single-sector innovators face.

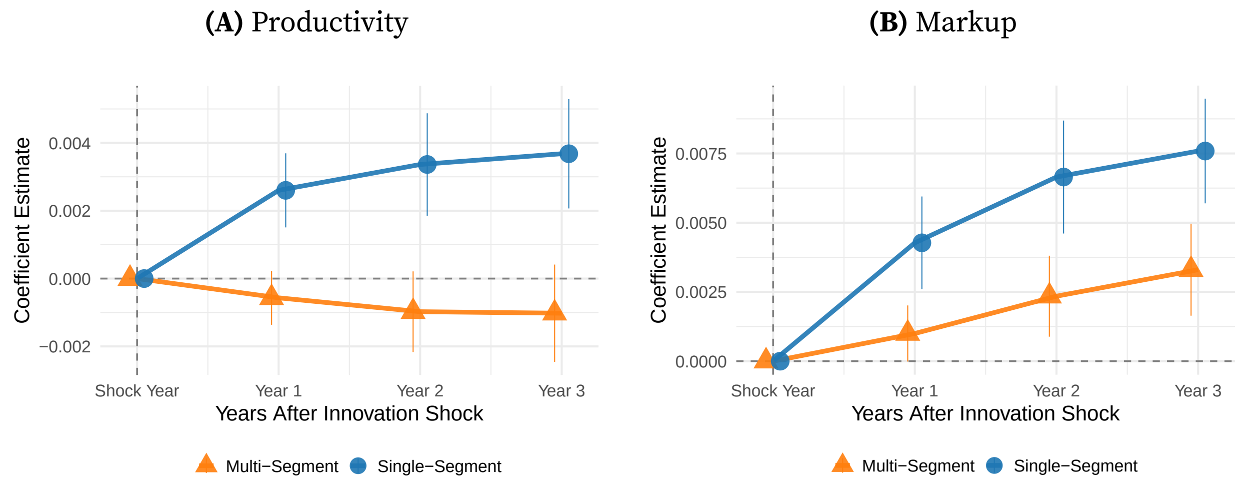

Figure 3. Productivity and Markup Responses to Innovation Shocks

Note: The innovation shock is measured as $\log(1 + V_{it})$, preserving firms with zero patent grants in a given year. Outcome variables and controls are measured in $\log(\cdot)$. The sample excludes utilities and finance sectors, as well as firms with missing or non-positive R&D, SG&A, employment, and sales. All variables cover the period 1990–2019. Vertical bars denote 95% confidence intervals based on standard errors clustered at the industry-year level.

The heterogeneity between firm types extends to productivity and markups, the margins that most directly capture the appropriability of innovation rents. Single-sector firms display a strong and monotonically increasing productivity response following an innovation shock, with the coefficient rising steadily from the shock year through year three (Figure 3A). Multi-sector firms, by contrast, exhibit a response that remains flat and statistically indistinguishable from zero throughout the entire horizon, with the gap between the two firm types widening persistently over time.

A parallel pattern emerges for markups, where the scope gap is present but more moderate (Figure 3B). Single-sector firms command significantly higher price-cost margins following an innovation shock, with the markup response growing monotonically and precisely estimated at all horizons. Multi-sector firms also exhibit a positive and increasing markup response, but the magnitude is substantially smaller throughout the horizon, and the gap between the two firm types widens persistently from year one onward. Taken together, both panels point to the same conclusion: firm scope systematically attenuates the returns to innovation. Single-sector firms translate innovation shocks into lasting gains in productive efficiency and market power, while multi-sector firms appropriate substantially smaller rents from equivalent shocks. This finding is consistent with Autor *et al.* (2020), in which higher-productivity firms command higher markups, and extends it by showing that scope moderates the productivity-markup nexus.[22]

Taken together, these results point to a coherent set of regularities, summarized by the following three stylized facts.[23]

Fact 1. Firm scope attenuates the transferable intangible investment response to innovation shocks: single-sector firms exhibit a stronger and more persistent reallocation toward transferable intangibles than their multi-sector counterparts

Fact 2. Innovation diffuses outward in focused firms. Following a shock, single-sector firms face a measurable increase in competitive threat from rivals, while multi-sector firms exhibit no significant competitive response at any horizon.

Fact 3. The gains from innovation in both productivity and markups are concentrated in single-sector firms. Multi-sector firms display significantly weaker responses along both margins, indicating that scope limits the appropriability of innovation rents.

4. Quantitative Analysis

4.1. Calibration and Identification

The model is calibrated to match moments that characterize how firm behavior and market outcomes vary systematically with scope. The state space is three-dimensional, so each moment at scope level n is constructed by aggregating over m and k dimensions.

[22] Appendix Figure A1 replicates the analysis using productivity and markup estimates based on Ackerberg *et al.* (2015) and Gandhi *et al.* (2020)

[23] Appendix Table A4 confirms that these patterns hold under a two-way fixed effects specification, providing a complementary check on the baseline local projection results.

For each scope level n, distribution given

$$\mu(n) = \sum_m \sum_k \mu(m,k,n).$$

The competitive threat at scope n is the probability that an incumbent loses a given sector, conditional on a challenger's successful innovation, combining displacement by a rival superstar and by a fringe entrant:

$$CT(n) \;=\; \mathbb{E}_n[\mathbb{P}_{c>s'}(m,k,n) \;+\; \mathbb{I}_{f>s'}(k,n)]\,.$$

Investment ratio and the average markup at scope level n are

$$I(n) = \mathbb{E}_n\left[\frac{I^{\text{Ver}}(m,k,n) + I^{\text{Hor}}(m,k,n) + I^{f}(m,k,n)}{I^{\text{Emb}}(m,k,n)}\right], \quad \sigma(n) = \mathbb{E}_n\left[\frac{1-\varepsilon\,\phi(m,k,n)}{(1-\phi(m,k,n))\,\varepsilon}\right].$$

Finally, the growth rate and productivity[24] at scope level n are

$$g(n) = \mathbb{E}_n\Big[\ln\lambda\Big(z^{\text{Ver}}(m,k,n) + \mathbb{P}_{s\geq s'}\,z^{\text{Hor}}(m,k,n) + \mathbb{I}_{f>s'}\,Z^{f}(m,k,n)\Big)\Big], \quad Q(n) = e^{g(n)}.$$

The model is disciplined by 36 empirical moments, comprising 18 targeted and 18 untargeted moments across six scope levels. The calibration relies on 16 parameters, of which four are set externally. The time discount rate is fixed at $\rho = 0.05$.Following Akcigit and Kerr (2018), the curvature parameters governing transferable intangible investment costs are set to $\vartheta^{\text{Ver}} = \vartheta^{\text{Hor}} = \vartheta^{f} = 2.0$. The remaining 12 parameters

$$(\varepsilon,\, \lambda,\, \theta,\, \alpha,\, \gamma,\, \beta,\, \xi,\, \gamma^{\text{Ver}},\, \gamma^{\text{Hor}},\, \gamma^{f},\, \gamma^{\text{Emb}},\, \vartheta^{\text{Emb}})$$

are estimated jointly by minimizing the distance between the model and empirical mo-

[24] Productivity satisfies $\dot{Q}(n,t)/Q(n,t) = g(n)$, integrating to $Q(n,t) = Q(n,t_0)\,e^{g(n)(t-t_0)}$ up to an arbitrary positive constant absorbed into the initial condition. $Q(n)$ thus denotes the gross growth factor over a unit interval; the normalization affects only levels and has no bearing on equilibrium allocations or identification.

ments across scope levels.[25]

Table 1. Parameter Values

External Calibration			Internal Calibration		
Parameter	**Description**	**Value**	**Parameter**	**Description**	**Value**
ρ	Discount rate	0.050	ε	CES parameter	0.867
ϑ^{Ver}	Vertical invest. curvature	2.000	λ	Quality step size	1.131
ϑ^{Hor}	Horizontal invest. curvature	2.000	θ	Embedded step size	1.148
ϑ^{f}	Fringe invest. curvature	2.000	α	Managerial curvature	0.514
			γ	Managerial scale	0.883
			β	Brand curvature	0.174
			ξ	Brand share	0.365
			γ^{Ver}	Vertical cost scale	9.247
			γ^{Hor}	Horizontal cost scale	3.368
			γ^{f}	Fringe cost scale	215.901
			γ^{Emb}	Embedded cost scale	2.561
			ϑ^{Emb}	Embedded curvature	2.459

Note: The upper limit for the number of production lines $\bar{n}$ is set to 6, and the upper bounds for $\bar{m}$ and $\bar{k}$ are set to 7 and 9, respectively.

The CES parameter ε and the managerial quality parameters α and γ are jointly disciplined by the variation in markups across scope levels: ε governs the level of price-cost margins, while α and γ shape how markups decline with scope through the span-of-control friction on managerial productivity. The quality step size λ is identified from the growth rate moments, since conditional on the equilibrium innovation rates, the level of $g(n)$ pins down λ directly. The cost scale parameters γ^{Ver}, γ^{Hor}, γ^{f}, and γ^{Emb}, together with the embedded curvature ϑ^{Emb} and step size θ, are jointly disciplined by the stationary scope distribution and the investment ratio, as they govern the marginal cost of each innovation strategy and therefore the equilibrium allocation of firms across the state space. The parameter $1-\xi$ determines the share of the embedded intangible stock allocated to organizational capital and hence directly governs the incumbent's ability to deter entry, so it is identified from the competitive threat moment. The parameter β governs the curvature of brand value and therefore enters the investment ratio and markup moments without affecting competitive threat; it is accordingly identified from the nonlinearity of

[25] Appendix C.2 describes the solution and estimation algorithm.

these moments in the embedded intangible stock across scope levels.[26]

Convex costs make the returns to each intangible type individually decreasing, yet the two are supermodular: a higher embedded level raises the marginal return to quality improvement and vice versa.[27] This complementarity means that individually decreasing returns do not preclude the possibility of multiple equilibria when the state space is unbounded. The bounded state space $\mathcal{M} \times \mathcal{K} \times \mathcal{N}$ is therefore a necessary condition for uniqueness, ensuring that the system of value functions defined over each state (m, k, n) admits a unique fixed-point solution on the finite state space.

4.2. Model Performance

Figure 4 shows that the model matches the targeted moments closely across all scope levels. The model reproduces the overall monotone decline in markups with scope (Figure 4A), though it underestimates markups at $n = 4$ and $n = 5$, where the data exhibit a non-monotonic uptick that the model does not fully capture. The investment ratio (Figure 4B) is tracked well overall, with the model capturing the sharp fall from $n = 1$ to $n = 2$ and the subsequent flattening toward higher scope levels. The slight divergence at $n = 4$ reflects the non-monotonic behavior in the data at $n = 5$ and the average loss minimization inherent in the estimation, rather than a systematic misfit. The scope distribution (Figure 4C) is replicated almost exactly, matching both the pronounced concentration of superstar firms at narrow scope and the rapid thinning of the distribution as scope expands. The tight fit across all three dimensions indicates that the calibrated model successfully internalizes the core tradeoffs between intangible investment, managerial span of control, and competitive dynamics that drive the empirical patterns.

[26] Appendix C2 shows how the moments vary with key parameters.

[27] Figure C3 shows the joint behavior of the value function across all pairwise combinations of the state space: $V_s(m, k)$, $V_s(m, n)$, and $V_s(k, n)$.

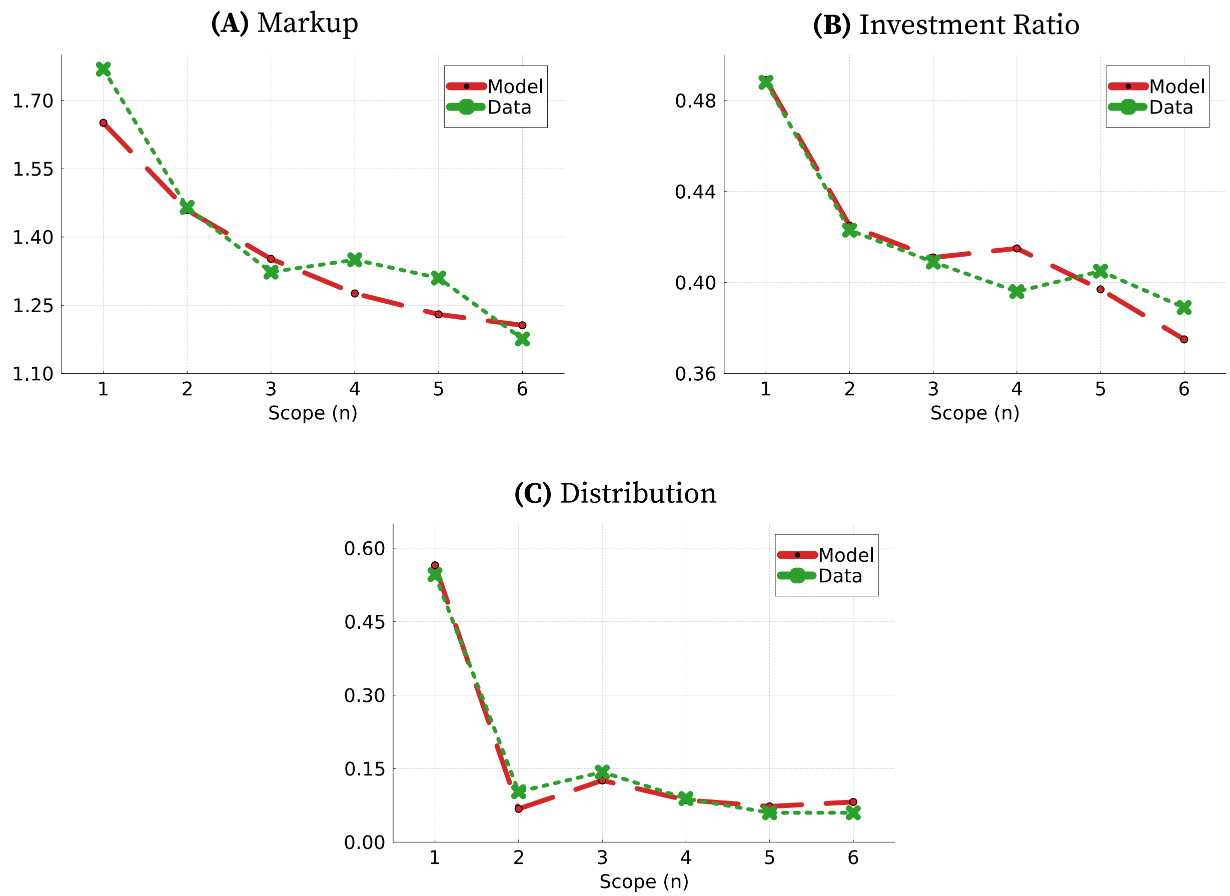

Figure 4. Targeted Moments: Markup, Investment Ratio and Distribution by Scope

Note: The green line represents the data and the red line the model along the balanced growth path. The sample excludes firms in the utilities and finance sectors, as well as firms with missing or non-positive R&D and SG&A expenditures. Markups, the investment ratio, and the size distribution are measured for the 2019 cross-section. The investment ratio is defined as transferable over embedded investment (see Section 3.2) and is winsorized at the 95th percentile; markups are winsorized at the 90th percentile.

Figure 5 examines model performance along untargeted dimensions. The model captures the overall declining trend in productivity with scope (Figure 5A), though it does not reproduce the non-monotonic behavior at intermediate scope levels present in the data. A similar pattern holds for growth rates (Figure 5B), where the model tracks the overall downward slope well but produces a smoother decline than the data. The competitive threat declines sharply from $n = 1$ and converges toward zero at high scope levels, with the model tracking the data closely throughout (Figure 5C). The close correspondence between model and data along untargeted dimensions suggests that the estimated parameters identify the structural mechanisms governing firm behavior across scope levels, rather than reflecting features specific to the targeted moment conditions.

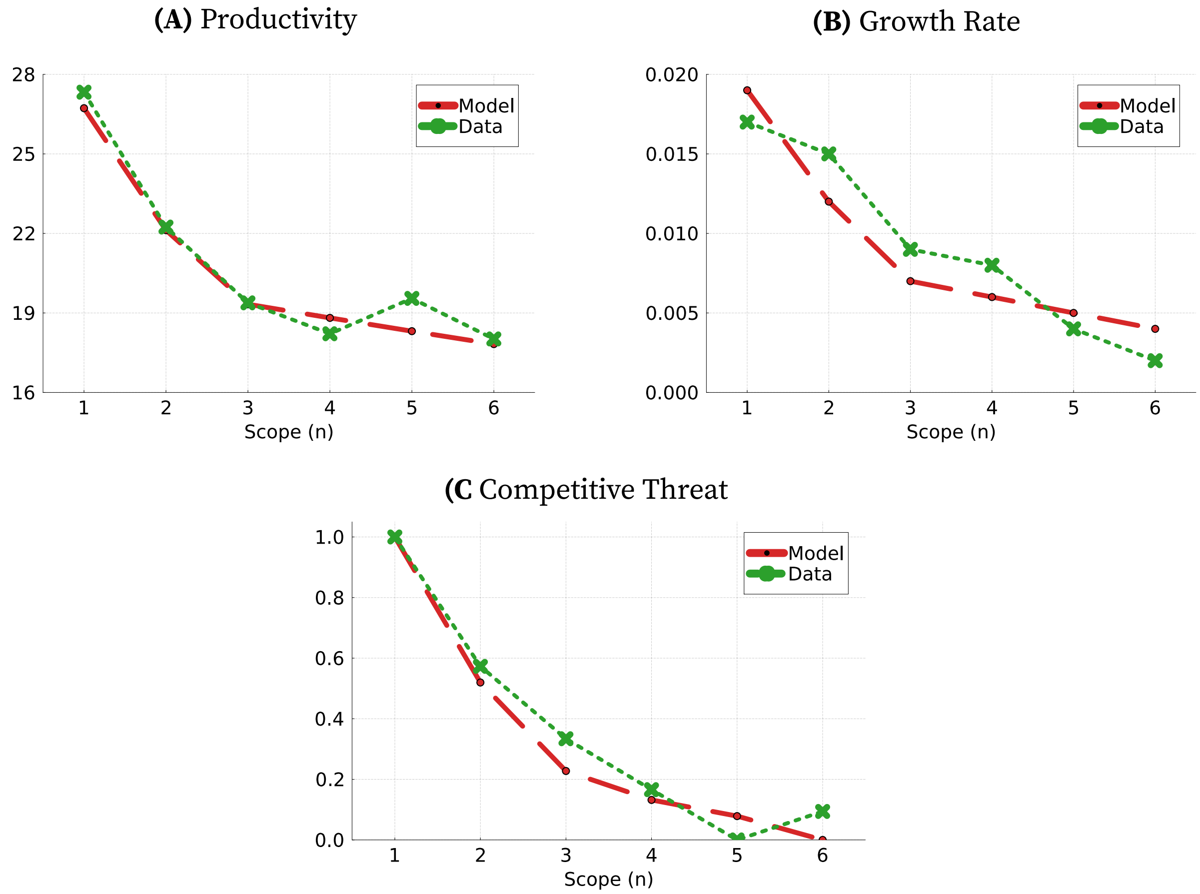

Figure 5. Untargeted Moments: Productivity, Growth Rate and Competitive Threat by Scope

Note: The green line represents the data and the red line the model along the balanced growth path. The sample excludes firms in the utilities and finance sectors, as well as firms with missing or non-positive R&D and SG&A expenditures. Productivity and competitive threat are measured for the 2019 cross-section and are described in Section 3.2. The growth rate is defined as the two-year log change in productivity, $\Delta_2 \ln(\text{prod})_{i,t} = \ln(\text{prod}_{i,t}) - \ln(\text{prod}_{i,t-1})$, averaged over 2016–2019 and winsorized at the 0.05th and 95th percentiles and productivity is winsorized at the 95th percentile. Competitive threat, in both the model and data, is computed as the average at each scope level and rescaled via min–max normalization, $(\bar{x}(n) - \min \bar{x}(n))/(\max \bar{x}(n) - \min \bar{x}(n))$, which lies in $[0, 1]$, with higher values indicating greater competitive pressure.

4.3. Superstar Firms Innovation Direction and Incentives

Figure 6 explains how the optimal innovation direction of superstar firms varies across scope, the transferable intangible gap m, and the embedded intangible level k. Two broad patterns emerge. First, scope itself exerts a first-order effect on innovation incentives: as firms expand from $n = 1$ to $n = 6$, the span-of-control constraint progressively erodes market share in each individual sector. Second, within each scope level, substantial heterogeneity persists depending on the composition of a firm's intangible endowment, and this relationship is highly nonlinear, particularly at low scope.

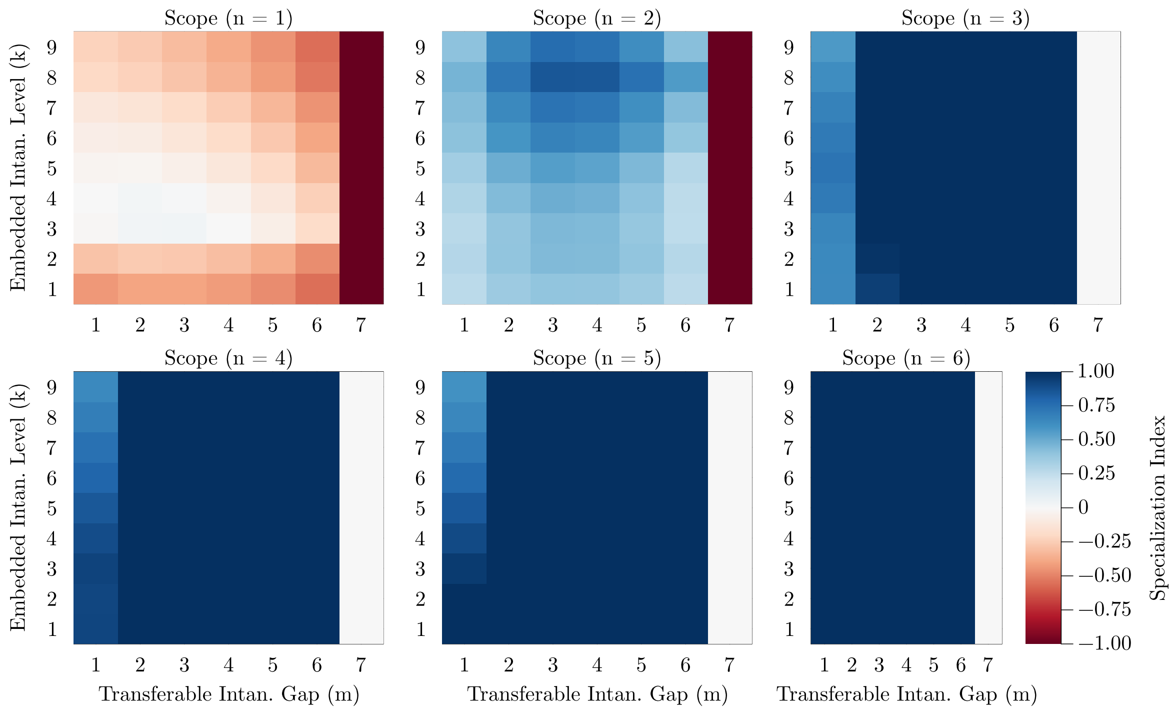

Figure 6. Superstar Firms Innovation Direction (Vertical vs Horizontal)

Each panel displays the specialization index $S(m,k) = \frac{z^{Ver}(m,k)-\mathbb{P}_{s>s'} z^{Hor}(m,k)}{z^{Ver}(m,k)+\mathbb{P}_{s>s'} z^{Hor}(m,k)}$ evaluated over the (m,k) grid for a given scope level n. The index lies in $[-1,1]$: blue cells indicate specialization toward vertical innovation, red cells indicate specialization toward horizontal innovation, and white cells indicate approximate indifference between the two directions. Each panel holds scope fixed and varies the transferable intangible gap m along the horizontal axis and the embedded intangible level k along the vertical axis, with the color gradient within a panel capturing how the innovation direction of superstar firms responds to their intangible stock composition at a given scale. Panels are arranged in order of increasing scope from $n = 1$ (top left) to $n = 6$ (bottom right).

At $n = 1$, firms with low embedded levels and moderate quality gaps optimally pursue horizontal innovation: without sufficient embedded stocks, cross-sector complementarities are weak and expanding scope offers higher returns than deepening quality within a single product line. As the embedded stock rises, however, the resulting increase in market share within the incumbent sector tilts incentives toward vertical investment. This pattern is further moderated by the transferable intangible gap: as m increases, the marginal return to vertical quality improvement declines, pushing firms back toward horizontal expansion, and at sufficiently high values of both m and k, the free mobility of embedded intangibles across sectors further reinforces the attractiveness of horizontal innovation, generating the non-monotonic pattern visible in the $n = 1$ panel. A similar logic governs $n = 2$, though the span-of-control constraint begins to bind more tightly: firms with high embedded stocks find that the market share dilution induced by scope expansion outweighs the gains from entering an additional sector, shifting them toward vertical innovation, while firms with lower embedded levels retain some horizontal incentives.

At the boundary $m = \bar{m}$ for low scope levels ($n = 1, 2$), the maximum transferable gap constitutes an absorbing state that concentrates incentives sharply toward horizontal innovation, reflected in the deep red cells along the rightmost column. This pattern changes markedly as scope expands: by $n \geq 3$, the rightmost column transitions to white, indicating that firms at the maximum quality gap neither improve vertically nor expand horizontally. At the boundary $m = \bar{m}$ under high scope, vertical improvement is precluded by the binding state space constraint, while the market share dilution from the span-of-control friction renders horizontal expansion unprofitable.

Taken together, Figure 6 establishes that innovation direction cannot be reduced to firm size alone. Scope shapes the overall level of innovation incentives by governing market share and managerial costs, while the composition of the intangible endowment determines the direction of investment within each scope level.

5. Counterfactual Analysis

5.1. Shutting Down Embedded Intangibles and Span of Control Constraint

To understand the effects of the span-of-control constraint and embedded intangibles on scope distribution, long-run economic growth, markups, and competitive threat, I consider two counterfactuals: shutting down the span-of-control constraint ($n^{\alpha} = 1$) and eliminating embedded intangibles ($\bar{k} = 1$).

Shutting down the span-of-control constraint allows superstar firms to expand scope without decreasing managerial quality per sector; consequently, the scope distribution shifts mass toward higher scope levels, since horizontal innovation no longer erodes managerial quality. Markups rise monotonically with scope, reversing the baseline pattern. The growth rate, in turn, exhibits an inverted-U pattern with scope.

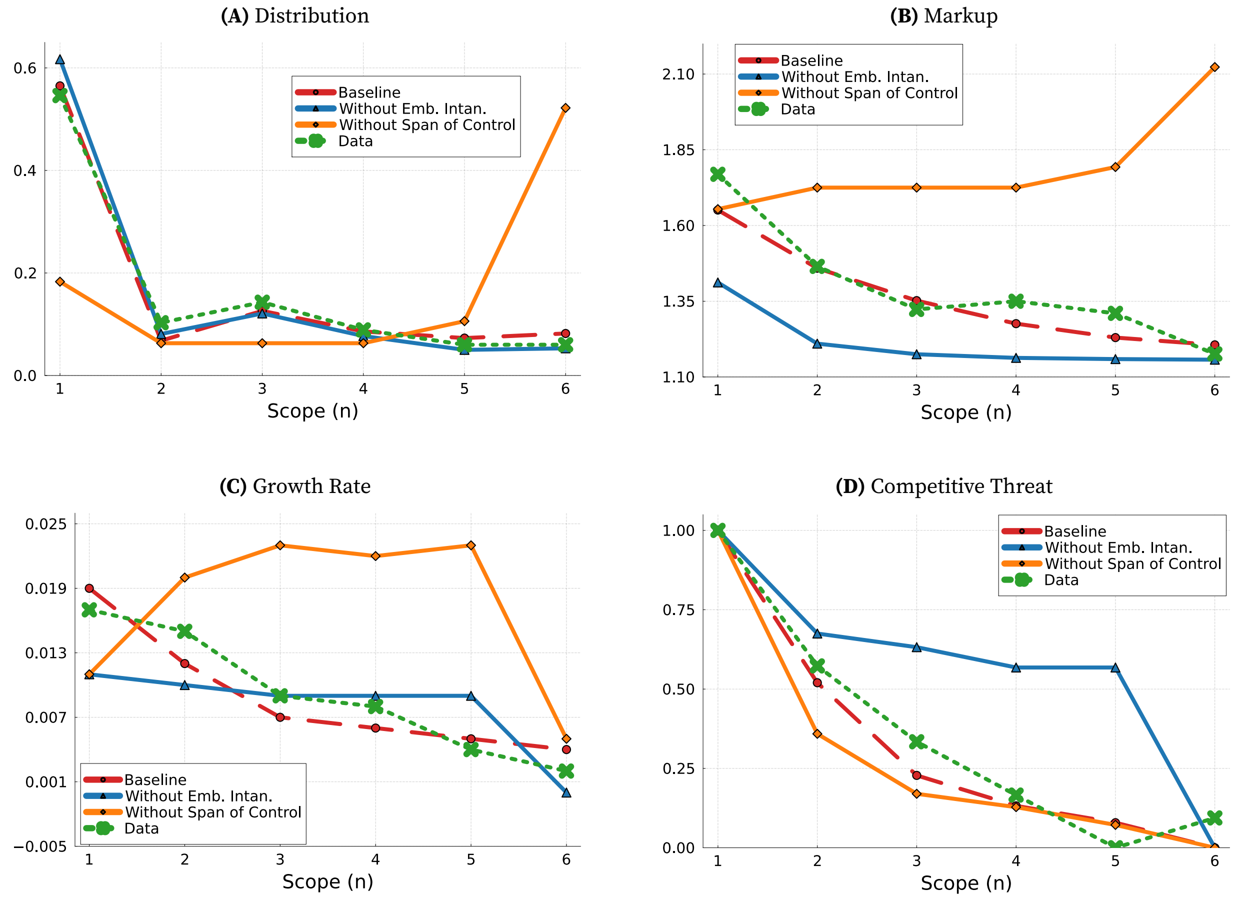

Figure 7. Counterfactual Analysis Under Different Cases

Note: The green line represents the data, whereas the red line shows the model results along the balanced growth path. The orange line corresponds to the case with the span-of-control channel shut down ($n^{\alpha} = 1$), while the blue line corresponds to the case with embedded intangibles shut down ($\bar{k} = 1$).Appendix **??** provides details on the calculation of the model moments. The reported competitive threat measure is computed as the average at each scope level and then rescaled using a min–max normalization, $(x - \min x)/(\max x - \min x)$, to lie in $[0, 1]$, capturing relative variation across scope levels, with higher values indicating greater competitive pressure.

At narrow scope, incumbent firms invest more intensively in innovation, driving growth above the baseline. As scope broadens, however, these same firms erect progressively higher intangible-based entry barriers, compressing competitive threat and attenuating innovation incentives at the frontier. Growth therefore peaks at intermediate scope levels and declines thereafter, though it remains above the baseline throughout—the sole exception being single-product firms, where growth falls below both the baseline and the data.

Eliminating embedded intangibles leaves the scope distribution largely unchanged, indicating that the span-of-control constraint is the primary determinant of the equilibrium allocation of firms across scope levels. The sharpest departure arises in the competitive threat: without the organizational capital component of embedded intangibles, incumbents can no longer erect intangible-based entry barriers, and competitive threat remains persistently elevated at scope levels where the baseline predicts near-zero entry. This divergence identifies embedded intangibles as the primary mechanism through which incumbent firms protect themselves from creative destruction as scope expands.

The two counterfactuals reveal distinct and complementary roles: the span-of-control constraint disciplines scope expansion, generates the declining markup gradient, and shapes the inverted-U growth profile across scope levels, while embedded intangibles govern the competitive threat profile and underwrite aggregate innovation. Neither channel alone is sufficient to account for the full set of empirical regularities documented in Section 3.3.

5.2. Shutting Down Entry Barriers and Misallocation

Misallocation in the model operates through two distinct channels: markup dispersion across firms, and the joint entry barrier of scope and embedded intangibles that prevents potential entrants from capturing a market unless their total managerial capacity exceeds the incumbent's. To quantify the first channel, I adapt Peters (2020) and decompose aggregate output as $Y_t = Q_t \times E_t \times M_t \times S_t$, where Q captures quality improvements, M captures misallocation from markup dispersion, and E and S are residual terms.[28] The misallocation term M is defined as

$$M_t = \frac{\exp\left(\sum_m \sum_k \sum_n \ln\left[\frac{\phi(m,k,n)}{\sigma(m,k,n)} + (1-\phi(m,k,n))\right]\mu(m,k,n)\right)}{\sum_m \sum_k \sum_n \left(\frac{\phi(m,k,n)}{\sigma(m,k,n)} + (1-\phi(m,k,n))\right)\mu(m,k,n)},$$

[28] For details see Appendix B.6.

namely the ratio of the geometric to the arithmetic mean of firm-level output shares, which endogenously evolves with markup dispersion.[29] To analyze the second channel, I conduct three counterfactual experiments that progressively dismantle the entry barriers embedded in the creative destruction conditions of the model.[30] The first removes only the scope component of the entry barrier, isolating the role of embedded intangibles in protecting incumbents. The second removes only the embedded intangible component, isolating the role of scope.[31]

Table 2. Misallocation and Growth Rate Under Different Entry Barrier Scenarios

	Baseline	No Scope Barrier (n)	No Embedded Barrier (k)
M	0.987	0.991	0.989
g	0.014	0.011	0.022

Table 2 presents the results. The baseline $M = 0.987$ confirms that markup dispersion alone generates only a modest output loss of around 1.3 percent, placing the primary source of misallocation in entry barriers rather than pricing distortions. Removing the scope barrier marginally improves M to 0.991 but lowers the growth rate from 0.014 to 0.011. The growth decomposition in Figure 8 reveals the mechanism: eliminating the scope barrier suppresses both vertical and horizontal innovation by incumbents, while the contribution of fringe firms rises substantially to become the dominant source of growth. This compositional shift is consistent with the inverted-U relationship between

[29]The term M differs slightly from Peters (2020): even with constant markups across superstar firms, dispersion between superstars and fringe firms persists through the market share ϕ.

[30]*Counterfactual 1 (No Scope Barrier).* The scope component $n^{1-\alpha}$ is shut down by setting $n_s = n_{s'} = 1$ in both conditions. The displacement conditions for the challenger superstar and fringe reduce to

$$\lambda\,(\theta^{k_s})^\alpha > (\theta^{k_{s'}})^\alpha \quad \text{and} \quad \lambda^{\bar{m}} > (\theta^{k_{s'}})^\alpha.$$

Counterfactual 2 (No Embedded Barrier). The embedded intangible component $(\theta^k)^\alpha$ is shut down by setting $k_s = k_{s'} = 1$ in both conditions,

$$\lambda\,(n_s)^{1-\alpha} > (n_{s'})^{1-\alpha} \quad \text{and} \quad \lambda^{\bar{m}} > (n_{s'})^{1-\alpha}.$$

[31]Eliminating both entry barriers simultaneously yields quantitatively similar results to removing the scope barrier alone and is therefore omitted from the analysis.

competition and innovation documented by Aghion *et al.* (2005): removing the scope barrier intensifies competitive pressure on incumbents beyond the point at which innovation incentives are maximized, eroding the managerial capacity advantage that sustains both vertical and horizontal investment and causing aggregate incumbent innovation to contract even as fringe innovation increases.

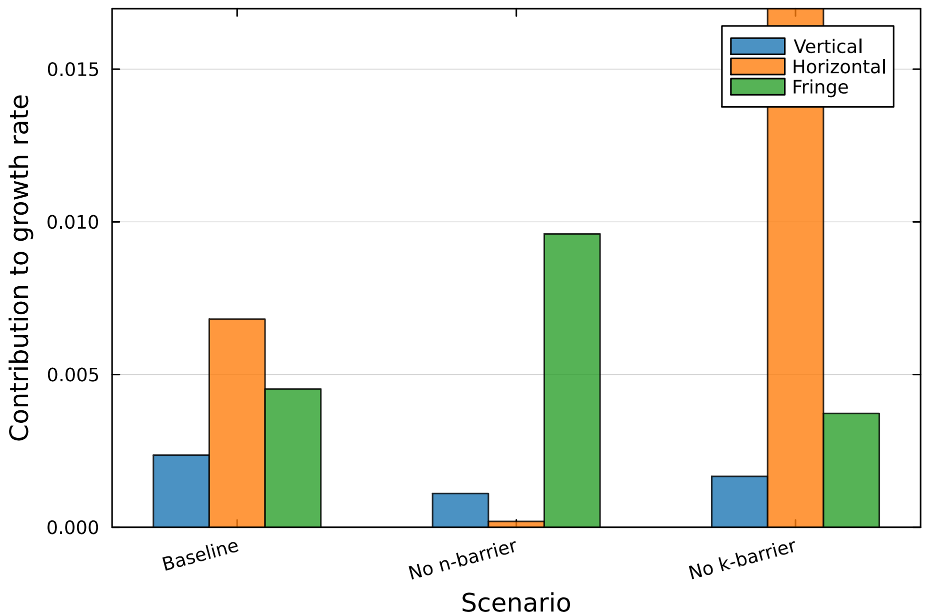

Figure 8. Growth Decomposition by Innovation Type and Entry Barrier

Note: The figure decomposes the aggregate growth rate into contributions from vertical innovation, horizontal innovation, and fringe entry across three scenarios: the baseline equilibrium, removal of the scope barrier (n-barrier), and removal of the embedded intangible barrier (k-barrier). Each bar reports the contribution of the respective innovation type to the balanced growth path rate g, as defined in equation (36).

The dominant finding is that eliminating the embedded intangible barrier raises the growth rate to 0.022 while improving M to 0.989, identifying the embedded intangible dimension as the quantitatively decisive source of misallocation. The growth decomposition reveals a strikingly different compositional pattern from the scope barrier case: horizontal innovation by incumbents expands dramatically and becomes the primary driver of aggregate growth, while the fringe contribution declines relative to the baseline. The mechanism is distinct from the competition channel: without the need to accumulate embedded stocks defensively, incumbents redeploy investment toward scope expansion, raising horizontal innovation rates and aggregate productivity growth simultaneously. This asymmetry between the two barriers reflects a fundamental difference in when and how they bind. The embedded barrier is particularly consequential for narrow-scope firms, since a challenger must match the incumbent's entire embedded stock to displace it, and this requirement is most demanding precisely when innovation incentives are strongest. The scope barrier, by contrast, constrains challengers only at later stages of horizontal expansion, when the span-of-control friction has already independently attenuated incumbent innovation incentives. The embedded barrier therefore operates as the binding constraint on entry and innovation at the margin where it matters most.[32]

6. Policy Implications

The counterfactual analysis in Section 5.2 establishes that the embedded intangible barrier is the quantitatively dominant source of misallocation, binding most severely for narrow-scope superstar firms where innovation incentives are strongest. Directly targeting embedded intangibles is not straightforward, however, since they are complementary to all three innovation margins: any instrument that depresses their accumulation simultaneously attenuates the vertical, horizontal, and embedded innovation incentives they underwrite. I therefore evaluate three progressive policy instruments — a scope-dependent profit levy τ_{scope}, a horizontal investment levy τ_{hor}, and an embedded intangible investment levy τ_{emb}. Each instrument is evaluated in isolation and combined with a flat subsidy to fringe entrants $\tau_f > 0$.[33] Following Berlingieri *et al.* (2025), I adopt a thresh-

[32]Figure C1 in the Appendix reports entry probabilities along the scope n and embedded intangible k dimensions for both challenger superstar and fringe firms.

[33]The three progressive instruments enter the model symmetrically. The scope-dependent profit levy modifies the operational profit flow in the superstar value function (26):

$$\left(1 - \tau_{\text{scope}}(n, j)\right) \pi(m, k, n),$$

where $\tau_{\text{scope}}(n, j) < 0$ constitutes a subsidy for firms with $n \leq j$ and $\tau_{\text{scope}}(n, j) > 0$ a tax for firms with $n > j$; since all innovation value differences are proportional to continuation values, the levy compresses or amplifies all three innovation margins simultaneously. The horizontal and embedded investment levies enter as multiplicative cost shifters in equation (8):

$$\left(1 + \tau_{\text{hor}}(n, j)\right) \gamma^{\text{Hor}} \left(z^{\text{Hor}}\right)^{\vartheta^{\text{Hor}}} \quad \text{and} \quad \left(1 + \tau_{\text{emb}}(k, j)\right) \gamma^{\text{Emb}} \left(z^{\text{Emb}}\right)^{\vartheta^{\text{Emb}}},$$

old approach that spans the full spectrum from a uniform tax to a uniform subsidy. For the scope-dependent instruments (τ_{scope} and τ_{hor}), the threshold j indexes firm scope: superstar firms with $n \leq j$ receive a subsidy of $+0.2$, while those with $n > j$ face a tax of -0.2. For the embedded intangible instrument (τ_{emb}), the threshold is instead defined over the embedded intangible level k: firms with $k \leq j$ receive a subsidy and those with $k > j$ face a tax. As the threshold j increases from its minimum (a uniform tax on all firms) to its maximum (a uniform subsidy), the policy transitions from penalizing all incumbents toward redistributing from wide-scope to narrow-scope firms. Net tax revenue is returned to households via lump-sum transfers when positive and financed by lump-sum levies when negative. Welfare is evaluated through a consumption-equivalence measure that quantifies the permanent percentage change in consumption leaving households indifferent between the baseline and the counterfactual allocation.[34]

Figure 9 reports the percentage change in the aggregate growth rate $\Delta_g(\%)$ across policy designs. The most striking pattern is the non-monotonic response of τ_{scope} in isolation: at low thresholds, a broad scope tax suppresses all three innovation margins through value compression and generates growth losses; as the threshold rises and the tax is increasingly concentrated on wide-scope incumbents, growth gains emerge and peak above 50% at intermediate values of j. Adding the fringe subsidy τ_f amplifies growth at every threshold and eliminates the losses at low thresholds: the combined $\tau_{\text{scope}} + \tau_f$ achieves the largest growth gains in the policy space, exceeding 60% at $j = 2$–3. This amplification arises because the fringe subsidy directly stimulates creative destruction while the scope tax redirects incumbent investment toward vertical quality improvement by penalizing the horizontal expansion margin. The horizontal and embedded investment taxes (τ_{hor} and τ_{emb}), when combined with τ_f, deliver stable positive growth gains of 20–30% across all thresholds, but their effect is attenuated relative to the scope tax because they do not

where $\tau < 0$ reduces costs (subsidy) and $\tau > 0$ raises them (tax), each acting directly on the targeted margin alone. The fringe subsidy $\tau_f > 0$ is strictly positive and reduces fringe investment costs in equation (27) to $(1 - \tau_f)\,\gamma^f(z_f)^{\vartheta^f}$. The first-order conditions (28)–(31) then imply that τ_{scope} shifts all three innovation rates through value compression or amplification depending on sign, τ_{hor} and τ_{emb} shift individual cost margins, and τ_f raises fringe innovation rates one-for-one.

[34] See Appendix B.7 for details.

compress the full value of incumbency and therefore generate a smaller reallocation of investment toward vertical innovation.

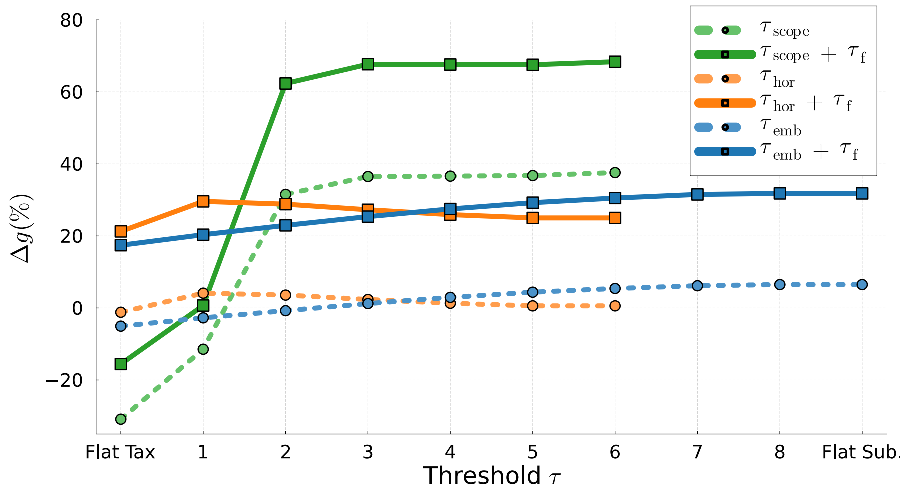

Figure 9. Flat and Progressive Policies Effect on Growth Rate

Note: Each line reports the percentage change in the aggregate growth rate $\Delta g(\%)$ relative to the baseline under alternative policy designs. Three policy instruments are considered: a scope tax τ_{scope}, a horizontal investment tax τ_{hor}, and an embedded intangible investment tax τ_{emb}, each applied in isolation (dashed lines) or combined with a flat subsidy to fringe firms τ_f (solid lines). The horizontal axis indexes the progressivity threshold τ: at the left endpoint (Flat Tax), all firms face a uniform tax regardless of scope; as τ increases, the tax applies only to firms with scope above the threshold, exempting smaller firms; at the right endpoint (Flat Sub.).Detailed thresholds and corresponding numerical results are reported in Appendix C1

Figure 10 reveals a sharp disconnect between growth maximization and welfare maximization. The policy that maximizes welfare is not the one that maximizes growth: setting the threshold at $j = 1$ combined with the fringe subsidy τ_f generates a welfare gain of approximately 15–20%, substantially exceeding all other configurations. By contrast, the policy designs that deliver the largest growth gains yield welfare gains that are considerably more modest or even negative. The mechanism arises from the convexity of investment costs: high-growth policies sustain elevated innovation rates only by pushing investment spending into the steeply rising region of the cost function. Because each additional unit of innovation flow requires a more than proportional increase in resources, the economy sacrifices a large share of current consumption to finance marginal increments in long-run productivity growth.

The intuition for the welfare dominance of the $j = 1$ design is precise and follows directly from the model's mechanisms. Subsidizing single-sector superstar firms activates all three innovation margins simultaneously: it raises the return to vertical quality improvement by increasing the profitability of the subsidized state, raises the return to embedded intangible accumulation since higher embedded stocks increase profits proportionally, and raises the return to horizontal expansion since firms that enter additional sectors cross the threshold into the taxed region only gradually.[35] Critically, this is exactly the margin where the misallocation identified in Section 5.2 binds most severely: narrow-scope firms face the highest embedded intangible barriers relative to their innovation incentives, and the subsidy corrects the wedge between private and social returns at precisely this margin. At the same time, taxing wide-scope incumbents collects revenue redistributed to households as lump-sum transfers, sustaining consumption alongside growth and avoiding the short-run consumption cost that drives the growth-welfare disconnect at higher thresholds.

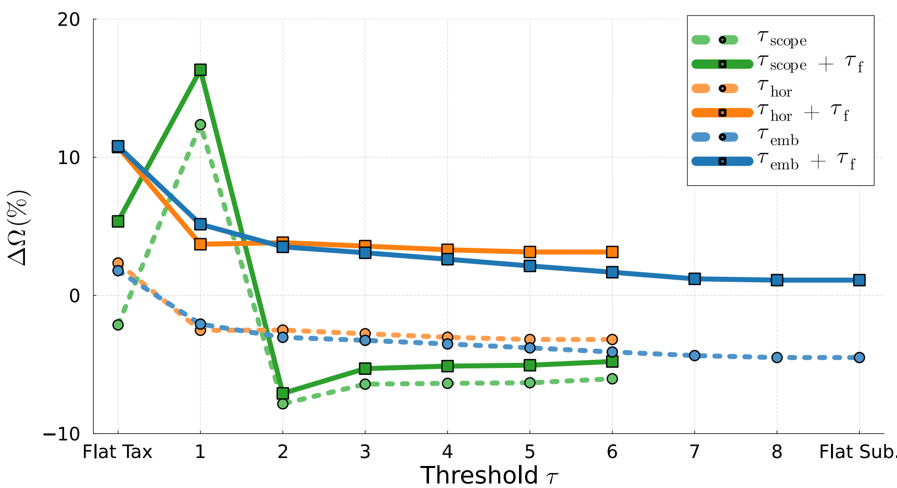

Figure 10. Flat and Progressive Policies Effect on Welfare

Note: Each line reports the percentage change in consumption equivalence welfare $\Delta\Omega(\%)$ relative to the baseline under alternative policy designs. Three policy instruments are considered: a scope tax τ_{scope}, a horizontal investment tax τ_{hor}, and an embedded intangible investment tax τ_{emb}, each applied in isolation (dashed lines) or combined with a flat subsidy to fringe firms τ_f (solid lines). The horizontal axis indexes the progressivity threshold τ: at the left endpoint (Flat Tax), all firms face a uniform tax regardless of scope; as τ increases, the tax applies only to firms with scope above the threshold, exempting smaller firms; at the right endpoint (Flat Sub.).Detailed thresholds and corresponding numerical results are reported in Appendix C1

The fringe subsidy τ_f is a necessary complement to any of the three incumbent-facing instruments: absent it, taxing superstar incumbents redistributes rents without improv-

[35]Formally, the subsidy at $n = 1$ raises $v(m, k, 1)$ and therefore increases $v(m_j, 1, k, 2) - v(m_j, k, 1)$, $v(m+1, k, 1) - v(m, k, 1)$, and $v(\mathbf{m}, k+1, 1) - v(\mathbf{m}, k, 1)$ simultaneously, expanding all three optimal innovation rates in equations (28)–(30).

ing the efficiency of creative destruction, generating lower gains across all threshold designs. This result reflects a fundamental asymmetry in the model. Incumbent-facing taxes reduce entry barriers by compressing the total managerial capacity advantage of existing superstars, but they do not restore fringe firms' ability to overcome those barriers. Since fringe firms constitute the fastest-growing segment of the economy, incentivizing their innovation rate generates aggregate gains disproportionate to the magnitude of the subsidy. Lowering the barrier without raising the innovation rate leaves most of the potential gains from creative destruction unrealized

7. Conclusion

This paper establishes a unified endogenous growth model in which the direction of innovative effort emerges endogenously from firms' expansion decisions. Horizontal expansion introduces span-of-control frictions that reduce managerial quality per sector and lower per-sector profitability. In response, multi-sector firms systematically reallocate investment away from transferable intangibles, which generate knowledge spillovers and fuel creative destruction, and toward embedded intangibles, which provide firm-specific competitive advantages. As scope and embedded intangible stock grow jointly, the incumbent superstar's total managerial capacity increases, raising entry barriers and suppressing creative destruction, which generates a wedge between private and social returns that instruments conditioned on firm size alone cannot correct. Correcting this wedge requires instruments that target firm scope and intangible composition directly, rather than relying on size-based thresholds, because size encompasses both vertical and horizontal dimensions and cannot capture intangible types of investment incentives. By doing so, policymakers can restore creative destruction and align private innovation incentives with social returns.

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Appendices

A. Empirical Appendix

A.1. Dataset

Table A1. Example Firms Segment in Compustat Segment Dataset

Company	Segments
TOYOTA MOTOR CORP	Financial Services Automotive All Other
PROCTER & GAMBLE CO	Health Care Grooming Corporate Beauty Baby, Feminine & Family Care Fabric & Home Care
TESLA INC	Energy Generation & Storage Automotive

A.2. Measurement Details

A.2.1. Production Function Estimation

Sample and specification. The estimation sample covers 1990–2019 and excludes the finance and utilities sectors. All estimations are conducted separately for each two-digit NAICS industry. The gross-output production function is specified as

$$y_{it} = \beta_l l_{it} + \beta_k k_{it} + \beta_m m_{it} + \omega_{it} + \epsilon_{it}, \tag{39}$$

where y_{it} is log gross output (`sale`), l_{it} is log labor (`emp`), k_{it} is log capital (`ppent`), and m_{it} is log intermediate inputs (`cogs`). The term ω_{it} denotes the unobserved firm-level productivity shock observed by the firm when making input decisions, and ϵ_{it} is an iid measurement error. The sample is restricted to observations satisfying

$$\texttt{sale} > 0, \quad \texttt{ppent} > 0, \quad \texttt{cogs} > 0, \quad \texttt{emp} > 0, \quad \texttt{capx} > 0, \quad \texttt{cogs} < \texttt{sale}, \tag{40}$$

where the final restriction ensures that intermediate input expenditure does not exceed total revenue.

All nominal variables are deflated to real terms prior to estimation. Output (`sale`) is deflated using the GDP deflator.[36] Intermediate inputs (`cogs`) is deflated using the Producer Price Index.[37] The capital stock (`ppent`) and capital expenditure (`capx`) are deflated using the Gross Private Domestic Investment deflator.[38]

Baseline: Olley and Pakes (1996) (OP). The baseline production function estimates follow OP, who resolve the simultaneity bias inherent in OLS estimation by exploiting the monotonicity of investment in unobserved productivity. Under the assumption that capital expenditure (`capx`) is strictly increasing in ω_{it} conditional on k_{it}, the productivity shock can be recovered as $\omega_{it} = h(k_{it}, \texttt{capx}_{it})$, which is approximated nonparametrically in the first stage.

Robustness: Ackerberg, Caves, and Frazer (2015) (ACF). The first robustness check follows ACF, who address the collinearity problem in the OP first stage by deferring identification of all input coefficients to the second stage, using the proxy inversion solely to control for unobserved productivity. The ACF estimator uses the same variable definitions as the baseline: `capx` as the proxy variable, with output, labor, capital, and intermediate inputs defined identically to the OP specification.

Robustness: Gandhi, Navarro, and Rivers (2020) (GNR). The second robustness check follows GNR, who identify the output elasticity of the flexible input nonparametrically from the firm's static optimality condition, without imposing functional form restrictions

[36] GDP Implicit Price Deflator: https://fred.stlouisfed.org/series/A191RD3A086NBEA.

[37] Producer Price Index by Commodity: https://fred.stlouisfed.org/series/WPUID61.

[38] Gross Private Domestic Investment Implicit Price Deflator: https://fred.stlouisfed.org/series/A006RD3A086NBEA.

on the production function. In this framework, the proxy variable is the intermediate input share $\frac{m_{it}}{y_{it}}$ (`cogs/sale`), which under the firm's cost-minimization condition is a sufficient statistic for productivity. The GNR estimator thus relaxes the parametric assumptions imposed by both OP and ACF, providing a more flexible benchmark for assessing the robustness of the output elasticity estimates.

A.2.2. Markup Estimation

Following De Loecker *et al.* (2020), firm-level markups are recovered from the firm's cost-minimization condition for the variable input. For firm i in year t, the markup is defined as

$$\mu_{it} = \frac{\theta_{it}^m}{\alpha_{it}^m}, \tag{41}$$

where θ_{it}^m is the output elasticity of the variable input (intermediate inputs, `cogs`), recovered from the first-stage production function estimation, and $\alpha_{it}^m = \frac{P_{it}^m M_{it}}{P_{it} Y_{it}}$ is the expenditure share of intermediate inputs in total revenue. This approach requires no assumption on demand or market structure and yields a firm-year-specific markup that captures heterogeneity in both technology and market power. Markup estimates below 0.5 and above 10 are excluded as economically inadmissible, being inconsistent with profit-maximizing behavior and likely reflecting numerical instability in the cost share estimation. This restriction affects approximately 0.75% of the estimation sample.

A.3. Additional Figures and Empirical Results

Table A2. Summary Statistics

Panel A: Compustat Fundamentals *Obs.: 71,139* *Firms: 8,125*

Variable	Mean	SD	P25	Median	P75
Sale	3,398,985	17,129,640	22,203	121,055	846,755
Emp	9,814	37,290	135	597	3,702
AT	4,502,256	23,855,832	26,457	142,066	981,467
XRD	147,233	746,922	1,589	7,513	35,977
XSGA	641,985	2,651,924	11,640	44,070	208,185
Productivity (OLS)	20.416	23.416	7.861	11.106	24.321
Markup (OLS)	1.543	1.042	0.988	1.210	1.665
Productivity (ACF)	30.191	49.551	6.410	8.391	36.216
Markup (ACF)	1.275	0.768	0.885	1.080	1.417
Productivity (GNR)	11.888	22.762	0.880	1.562	7.911
Markup (GNR)	1.291	0.914	0.826	1.014	1.382

Panel B: Compustat–Segment Merged (including ADR) *Obs.: 43,007* *Firms: 5,830*

Variable	Mean	SD	P25	Median	P75
Sale	3,597,348	16,753,747	29,041	166,258	1,120,375
Emp	10,466	37,694	154	708	4,482
AT	5,101,380	26,098,750	37,743	211,989	1,398,987
XRD	180,872	867,769	2,154	10,697	49,903
XSGA	721,384	2,891,804	14,998	59,082	266,119
Productivity (OLS)	21.055	24.665	8.054	11.264	25.035
Markup (OLS)	1.549	1.056	0.986	1.205	1.674
Productivity (ACF)	31.702	53.825	6.469	8.471	36.590
Markup (ACF)	1.287	0.785	0.889	1.082	1.423
Productivity (GNR)	11.991	23.298	0.891	1.549	7.952
Markup (GNR)	1.300	0.935	0.825	1.011	1.388
Market Fluidity	6.476	3.120	4.203	6.020	8.183

Panel C: Compustat–Segment Merged (excluding ADR) *Obs.: 35,964* *Firms: 4,905*

Variable	Mean	SD	P25	Median	P75
Sale	2,171,465	10,468,548	25,104	135,367	826,899
Emp	6,440	23,946	133	571	3,212
AT	3,083,646	19,749,630	30,984	167,030	977,718
XRD	127,011	736,443	1,813	8,868	39,625
XSGA	476,449	2,256,258	13,128	50,168	208,927
Productivity (OLS)	20.556	24.630	7.890	11.082	23.682
Markup (OLS)	1.545	1.048	0.986	1.207	1.673
Productivity (ACF)	31.199	53.967	6.398	8.439	34.253
Markup (ACF)	1.295	0.795	0.895	1.089	1.434
Productivity (GNR)	11.672	23.026	0.883	1.473	7.212
Markup (GNR)	1.298	0.928	0.826	1.014	1.391
Market Fluidity	6.451	3.104	4.189	6.005	8.156

Notes: This table reports summary statistics for the three estimation samples. Panel A covers Compustat Fundamentals Annual restricted to firm-years with positive sales, employment, R&D (XRD), and SG&A (XSGA). Panel B merges Panel A with the Compustat Segment database, retaining all countries. Panel C restricts Panel B to U.S.-incorporated firms (*fic* = USA).

Table A3. Sector-Level Elasticities: OP, ACF, and GNR

		OP				ACF				GNR			
NAICS	Sector	β_M	β_K	β_L	RTS	β_M	β_K	β_L	RTS	β_M	β_K	β_L	RTS
11	Agriculture, Forestry & Fishing	0.834	0.032	0.100	0.966	0.612	0.146	0.137	0.894	0.676	0.209	0.148	1.030
21	Mining & Oil and Gas Extraction	0.719	0.118	0.050	0.887	0.692	0.280	0.001	0.972	0.500	0.386	0.111	0.997
23	Construction	0.931	0.023	0.012	0.966	—	—	—	—	0.826	0.018	0.131	0.975
31	Food & Beverage Manufacturing	0.853	0.029	0.066	0.947	—	—	—	—	0.659	0.148	0.189	0.996
32	Chemical & Paper Manufacturing	0.612	0.019	0.275	0.907	0.351	0.094	0.501	0.946	0.489	0.106	0.399	0.994
33	Machinery & Computer Manufacturing	0.724	0.027	0.184	0.935	0.692	0.076	0.199	0.967	0.617	0.091	0.315	1.020
42	Wholesale Trade	0.832	0.046	0.086	0.963	0.504	0.065	0.330	0.900	0.790	0.054	0.160	1.000
44	Retail Trade I	0.782	0.043	0.106	0.931	—	—	—	—	0.728	0.120	0.203	1.050
45	Retail Trade II	0.780	0.046	0.064	0.889	0.466	0.094	0.290	0.850	0.681	0.154	0.207	1.040
48	Transportation & Warehousing	—	—	—	—	0.628	0.120	0.113	0.860	0.755	0.196	0.065	1.020
51	Information	0.490	0.059	0.336	0.884	—	—	—	—	0.354	0.128	0.468	0.951
53	Real Estate & Rental	—	—	—	—	—	—	—	—	0.363	0.188	0.306	0.857
54	Professional & Technical Services	0.634	0.054	0.183	0.871	0.119	0.175	0.583	0.877	0.580	0.132	0.310	1.020
56	Administrative & Support Services	0.766	0.053	0.078	0.897	—	—	—	—	0.655	0.170	0.185	1.010
61	Educational Services	0.708	0.109	0.043	0.860	0.446	0.137	0.281	0.864	0.486	0.211	0.287	0.984
62	Health Care & Social Assistance	0.819	0.052	0.011	0.882	0.396	0.122	0.317	0.834	0.708	0.110	0.117	0.935
71	Arts, Entertainment & Recreation	0.871	0.054	0.023	0.948	0.424	0.246	0.177	0.847	0.649	0.179	0.135	0.963
72	Accommodation & Food Services	0.819	0.095	0.021	0.935	0.548	0.201	0.185	0.934	0.739	0.197	0.110	1.040
81	Other Services	—	—	—	—	0.796	0.053	0.105	0.954	0.665	0.102	0.139	0.905
99	Nonclassifiable Establishments	0.748	0.007	0.159	0.915	0.418	0.084	0.393	0.894	0.617	0.074	0.251	0.942

Notes: β_M, β_K, and β_L denote the estimated output elasticities with respect to materials (flexible input), capital, and labor (both dynamic inputs), respectively. RTS $\equiv \beta_M + \beta_K + \beta_L$ is the implied returns to scale. OP refers to olleypakes1996; ACF refers to ackerberg2015identification; GNR refers to gandhi2020identification. Missing values (—) indicate that the estimation did not produce valid positive elasticities or RTS $\notin [0.8, 1.2]$.

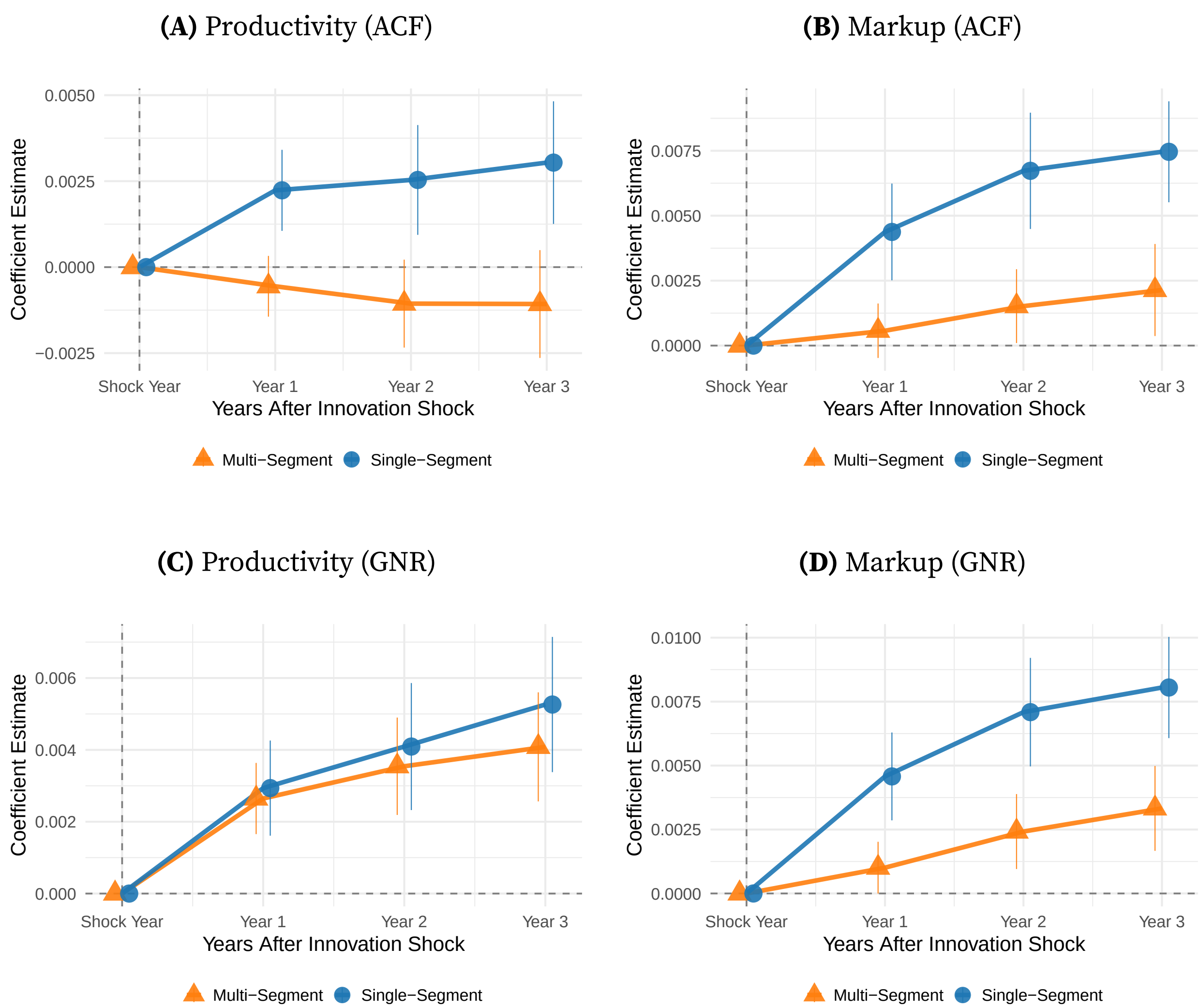


Figure A1. Productivity and Markup Responses to Innovation Shocks: ACF and GNR

Note: Panels (A) and (B) report estimates based on Ackerberg *et al.* (2015); Panels (C) and (D) report estimates based on Gandhi *et al.* (2020). The innovation shock is measured as $\log(1+\text{PatentValue}_{it})$, preserving firms with zero patent grants in a given year. Outcome variables and controls are measured in $\log(\cdot)$. The sample excludes utilities and finance sectors, as well as firms with missing or non-positive R&D, SG&A, employment, and sales. All variables cover the period 1990–2019. Vertical bars denote 95% confidence intervals based on standard errors clustered at the industry-year level.

Table A4. Regression Results

Panel A: *Markup* and *Productivity*

	Pooled OLS		Two-way FE	
	(2) *Markup*	(3) *Productivity*	(6) *Markup*	(7) *Productivity*
Production Lines	-0.038^{***}	-0.056^{***}	-0.041^{***}	-0.044^{***}
	(0.004)	(0.008)	(0.004)	(0.003)
Num. Obs.	40,008	40,008	40,008	40,008
Adj. R^2	0.268	0.197	0.279	0.833
Covariates	Yes	Yes	Yes	Yes
FE: *Year*	No	No	Yes	Yes
FE: *Industry*	No	No	Yes	Yes

Panel B: *Investment Ratio* and *Fluidity*

	Pooled OLS		Two-way FE	
	(1) *Transferable/Embedded*	(4) *Comp. Threat*	(5) *Transferable/Embedded*	(8) *Comp. Threat*
Production Lines	-0.011^{***}	-0.037^{***}	-0.014^{***}	-0.017^{**}
	(0.002)	(0.006)	(0.002)	(0.006)
Num. Obs.	42,831	31,318	42,831	31,318
Adj. R^2	0.977	0.208	0.978	0.305
Covariates	Yes	Yes	Yes	Yes
FE: *Year*	No	No	Yes	Yes
FE: *Industry*	No	No	Yes	Yes

Notes: Each column reports coefficients from a separate regression. Standard errors are clustered by *firm id* (gvkey) in parentheses. Significance levels: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. Columns 1–4: Pooled OLS specifications; Columns 5–8: Two-way fixed effects (year and industry). Covariates include sale, xrd, and xsga.

B. Model Appendix

B.1. Final Good Sector Demand

The final good sector's profit maximization problem is:

$$\max_{y_{jt}} \exp\left(\int_0^1 \ln y_{jt} dj\right) - \int_0^1 p_{jt} y_{jt} dj. \tag{42}$$

The first-order condition yields the inverse demand function for each intermediate good j:

$$p_{jt} = \frac{Y_t}{y_{jt}}. \tag{43}$$

B.2. Intermediate Good Sector Demand

The cost minimization problem for the intermediate sector j is:

$$\min_{y_{sjt}, y_{fjt}} \quad p_{sjt} y_{sjt} + p_{fjt} y_{fjt} \quad s.t. \quad y_{jt} = \left(\Xi(e_s) y_{sjt}^{\varepsilon} + (1 - \Xi(e_s)) y_{fjt}^{\varepsilon}\right)^{\frac{1}{\varepsilon}}. \tag{44}$$

The first-order condition with respect to the superstar firm's output y_{sjt} is:

$$p_{sjt} = \lambda\, \Xi(e_s) y_{jt}^{1-\varepsilon} y_{sjt}^{\varepsilon - 1}, \tag{45}$$

where λ is the Lagrange multiplier. Raising both sides to the power of $\frac{\varepsilon}{\varepsilon-1}$ and simplifying allows to solve for the ideal price index p_j for sector j:

$$\lambda = \left(\Xi(e_s)^{\frac{-1}{\varepsilon-1}} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} + (1 - \Xi(e_s))^{\frac{-1}{\varepsilon-1}} p_{fjt}^{\frac{\varepsilon}{\varepsilon-1}}\right)^{\frac{\varepsilon-1}{\varepsilon}} \equiv p_j. \tag{46}$$

Substituting $\lambda = p_j$ back into the first-order condition yields the inverse demand function faced by the superstar firm:

$$p_{sjt} = p_j\, \Xi(e_s) y_{jt}^{1-\varepsilon} y_{sjt}^{\varepsilon-1}. \tag{47}$$

Finally, substituting the final good producer's demand $y_{jt} = Y_t/p_j$ and solving for y_{sjt} provides demand function:

$$y_{sjt} = p_j^{\frac{\varepsilon}{1-\varepsilon}}\,\Xi(e_s)^{\frac{1}{1-\varepsilon}} p_{sjt}^{\frac{1}{\varepsilon-1}} Y_t. \tag{48}$$

B.3. Superstar Firm Maximization Problem: Bertrand Competition

The superstar firm competes à la Bertrand with a continuum of fringe firms. Its profit maximization problem in industry j is:

$$\max_{p_{sjt}} \left(p_{sjt} - MC_{sjt}\right) y_{sjt} \quad s.t. \quad y_{sjt} = p_j^{\frac{\varepsilon}{1-\varepsilon}}\,\Xi(e_s)^{\frac{1}{1-\varepsilon}} p_{sjt}^{\frac{1}{\varepsilon-1}} Y_t. \tag{49}$$

Substituting the expression for p_j into the demand function, the objective function can be expanded as:

$$\max_{p_{sjt}} \left[p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}}\,\Xi(e_s)^{\frac{1}{1-\varepsilon}} Y_t \left(\Xi(e_s)^{\frac{-1}{\varepsilon-1}} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} + (1-\Xi(e_s)) p_{fjt}^{\frac{\varepsilon}{\varepsilon-1}}\right)^{-1} \right. \tag{50}$$

$$\left. - MC_{sjt} \left(\Xi(e_s)^{\frac{-1}{\varepsilon-1}} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} + (1-\Xi(e_s)) p_{fjt}^{\frac{\varepsilon}{\varepsilon-1}}\right)^{-1} p_{sjt}^{\frac{1}{\varepsilon-1}}\,\Xi(e_s)^{\frac{1}{1-\varepsilon}} Y_t \right]. \tag{51}$$

After computing the derivative and factoring common terms, this condition can be expressed as:

$$\begin{aligned}\frac{\partial \pi}{\partial p_{sjt}} = {} & Y_t\,\Xi(e_s)^{\frac{1}{1-\varepsilon}} \left(\Xi(e_s)^{\frac{-1}{\varepsilon-1}} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} + (1-\Xi(e_s))\, p_{fjt}^{\frac{\varepsilon}{\varepsilon-1}}\right)^{-1} \times \left\{ \left[\frac{\varepsilon}{\varepsilon-1} p_{sjt}^{\frac{1}{\varepsilon-1}} - MC_{sjt} \frac{1}{\varepsilon-1} p_{sjt}^{\frac{2-\varepsilon}{\varepsilon-1}}\right] \right. \\ & \left. - \left(p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} - MC_{sjt}\, p_{sjt}^{\frac{1}{\varepsilon-1}}\right) \left(\Xi(e_s)^{\frac{-1}{\varepsilon-1}} \frac{\varepsilon}{\varepsilon-1} p_{sjt}^{\frac{1}{\varepsilon-1}} \left(\Xi(e_s)^{\frac{-1}{\varepsilon-1}} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} + (1-\Xi(e_s))\, p_{fjt}^{\frac{\varepsilon}{\varepsilon-1}}\right)^{-1}\right) \right\}.\end{aligned} \tag{52}$$

$$0 = \left[\frac{\varepsilon}{\varepsilon - 1} p_{sjt}^{\frac{1}{\varepsilon-1}} - MC_{sjt} \frac{1}{\varepsilon - 1} p_{sjt}^{\frac{2-\varepsilon}{\varepsilon-1}} \right]$$
$$- \left(p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} - MC_{sjt} \, p_{sjt}^{\frac{1}{\varepsilon-1}} \right) \left(\Xi(e_s)^{\frac{-1}{\varepsilon-1}} \frac{\varepsilon}{\varepsilon - 1} p_{sjt}^{\frac{1}{\varepsilon-1}} \left(\Xi(e_s)^{\frac{-1}{\varepsilon-1}} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} + (1 - \Xi(e_s)) \, p_{fjt}^{\frac{\varepsilon}{\varepsilon-1}} \right)^{-1} \right). \tag{53}$$

To simplify, multiply both sides by p_{sjt} and substitute the market share definition ϕ_{sjt} from (19), yielding:

$$\left(p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} - \text{MC}_{sjt} p_{sjt}^{\frac{1}{\varepsilon-1}} \right) \left(\phi_{sjt} \frac{\varepsilon}{\varepsilon - 1} \right) = \left[\frac{\varepsilon}{\varepsilon - 1} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} - \text{MC}_{sjt} \frac{1}{\varepsilon - 1} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}} \right]. \tag{54}$$

First, divide both sides by $p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}}$, then by $\frac{\varepsilon}{\varepsilon-1}$. After rearranging terms, the expression simplifies to:

$$\varepsilon(1 - \phi_{sjt}) = \frac{\text{MC}_{sjt}}{p_{sjt}} (1 - \varepsilon \phi_{sjt}). \tag{55}$$

Solving for the optimal price p_{sjt} gives:

$$p_{sjt} = \frac{1 - \varepsilon \phi_{sjt}}{\varepsilon(1 - \phi_{sjt})} \cdot \text{MC}_{sjt}. \tag{56}$$

To determine the optimal labor demand, equating output (6) and demand (48) yields

$$q_{sjt} \, \psi(e_s, n_s) \, l_{sjt} = p_j^{\frac{\varepsilon}{1-\varepsilon}} \chi(e_s)^{\frac{1}{1-\varepsilon}} p_{sjt}^{\frac{1}{\varepsilon-1}} Y_t. \tag{57}$$

Multiplying both sides by p_{sjt} and dividing by w_t gives

$$p_{sjt} \underbrace{\frac{q_{sjt} \, \psi(e_s, n_s)}{w_t}}_{\text{inverse MC}_{sjt}} l_{sjt} = \underbrace{p_j^{\frac{\varepsilon}{1-\varepsilon}} \chi(e_s)^{\frac{1}{1-\varepsilon}} p_{sjt}^{\frac{\varepsilon}{\varepsilon-1}}}_{\phi_{sjt}} \underbrace{\frac{Y_t}{w_t}}_{\omega_t^{-1}}. \tag{58}$$

This expression leads directly to equation (24).

B.4. Superstar Firm Maximization Problem: Cournot Competition

In the à la Cournot setup, a superstar firm and a continuum of fringe firms compete by choosing quantities to sell rather than engaging in price competition as in the à la

Bertrand case. Its profit-maximization problem in industry j is

$$\max_{y_{sjt}} \pi_{sjt} = \max_{y_{sjt}} (p_{sjt} - MC_{sjt})\, y_{sjt}, \tag{59}$$

$$s.t. \quad p_{sjt} = \chi(e_s)\, y_{jt}^{-\varepsilon}\, y_{sjt}^{\varepsilon-1}\, Y_t, \quad \text{and} \quad y_{jt} = \left(\chi(e_s)\, y_{sjt}^{\varepsilon} + y_{fjt}^{\varepsilon}\right)^{1/\varepsilon}. \tag{60}$$

Differentiate the profit function with respect to y_{sjt}:

$$\frac{d\pi_{sjt}}{dy_{sjt}} = (p_{sjt} - MC_{sjt}) + y_{sjt}\frac{dp_{sjt}}{dy_{sjt}} = 0. \tag{61}$$

Differentiate the inverse demand (60) to obtain

$$\frac{dp_{sjt}}{dy_{sjt}} = p_{sjt}\left(-\varepsilon\frac{1}{y_{jt}}\frac{dy_{jt}}{dy_{sjt}} + \frac{\varepsilon-1}{y_{sjt}}\right). \tag{62}$$

Differentiate y_{jt} with respect to y_{sjt}:

$$\frac{dy_{jt}}{dy_{sjt}} = \chi(e_s)\, y_{sjt}^{\varepsilon-1}\, y_{jt}^{1-\varepsilon}. \tag{63}$$

Substituting (63) and (60) into (61) yields

$$(p_{sjt} - MC_{sjt}) + y_{sjt}\, p_{sjt}\left(-\varepsilon\chi(e_s)\, y_{sjt}^{\varepsilon-1} y_{jt}^{-\varepsilon} + \frac{\varepsilon-1}{y_{sjt}}\right) = 0. \tag{64}$$

Using the market-share definition

$$\frac{p_{sjt}\, y_{sjt}}{Y_t} = \chi(e_s)\, y_{sjt}^{\varepsilon}\, y_{jt}^{-\varepsilon},$$

and dividing both sides by p_{sjt}, rearrangement gives the inverse markup condition

$$\frac{MC_{sjt}}{p_{sjt}} = \varepsilon(1 - \phi_{sjt}). \tag{65}$$

The relative price ratio between fringe and superstar firms is therefore

$$\frac{p_{fjt}}{p_{sjt}} = \frac{1-\phi_{sjt}}{\phi_{sjt}} \frac{MC_{fjt}}{MC_{sjt}}. \tag{66}$$

Using the inverse demand expressions leads to

$$\chi(e_s) \left(\frac{y_{fjt}}{y_{sjt}}\right)^{\varepsilon-1} = \frac{1-\phi_{sjt}}{\phi_{sjt}} \frac{MC_{fjt}}{MC_{sjt}}. \tag{67}$$

Substituting the marginal-cost expressions for the fringe and superstar firms yields

$$\left(\frac{y_{fjt}}{y_{sjt}}\right)^{\varepsilon-1} = \frac{1}{\chi(e_s)} \frac{1-\phi_{sjt}}{\phi_{sjt}} \frac{1}{\lambda^{m_j}\,\psi(e_s, n_s)}. \tag{68}$$

Equation (68) shows that the relative output depends on the market share, the quality gap, the embedded intangible level, and the firm's production-line parameter. Rearranging the market-share definition gives

$$\chi(e_s)\, y_{sjt}^{\varepsilon}\, y_{jt}^{-\varepsilon} = \frac{\chi(e_s)\, y_{sjt}^{\varepsilon}}{\chi(e_s)\, y_{sjt}^{\varepsilon} + y_{fjt}^{\varepsilon}} = \frac{1}{1 + \dfrac{1}{\chi(e_s)} \left(\dfrac{y_{fjt}}{y_{sjt}}\right)^{\varepsilon}}, \tag{69}$$

which shows that market share depends on the output gap and the embedded intangible level. Therefore, the output gap depends on the quality gap, the embedded intangible level, and the number of production lines associated with superstar s.

B.5. Aggregate Output and Growth Rate

Using the superstar (6) and fringe firm (7) output equations into (3) gives

$$Y_t = \exp\left(\int_0^1 \ln\left[(\xi e_{st})^{\beta}\left(q_{sjt}\,\frac{((1-\xi)e_{st})^{\alpha}}{\gamma n_{st}^{\alpha_s}}\, l_{sjt}\right)^{\varepsilon} + \left(q_{fjt}\, l_{fst}\right)^{\varepsilon}\right]^{1/\varepsilon} dj\right)$$

$$Y_t = \exp\left(\int_0^1 \ln\left[\left(q_{sjt}^{\varepsilon}\left((\xi e_{st})^{\beta}\left(\frac{((1-\xi)e_{st})^{\alpha}}{\gamma n_{st}^{\alpha_s}}\, l_{sjt}\right)^{\varepsilon} + \left(\lambda^{-m_{jt}}\, l_{fst}\right)^{\varepsilon}\right)\right)\right]^{1/\varepsilon} dj\right)$$

$$= \underbrace{\exp\left(\int_0^1 \ln q_{sjt} dj\right)}_{Q_t} \exp\left(\int_0^1 \ln\left[((\xi e_{st})^{\beta}\left(\frac{((1-\xi)e_{st})^{\alpha}}{\gamma n_{st}^{\alpha_s}} l_{sjt}\right)^{\varepsilon} + \left(\lambda^{-m_{jt}} l_{fst}\right)^{\varepsilon}\right]^{\frac{1}{\varepsilon}} dj\right), \tag{70}$$

and , along with the labor demands in (24) and (**??**), and factoring out ω_t^{-1}, aggregate output can be expressed as

$$Y_t = Q_t \, \omega_t^{-1} \exp\left(\int_0^1 \ln\left[(\xi e_{st})^{\beta}\left(\frac{((1-\xi)e_{st})^{\alpha}}{\gamma n_{st}^{\alpha_s}} \frac{\phi_{sjt}}{\sigma_{sjt}}\right)^{\varepsilon} + \left(\lambda^{-m_{jt}}(1-\phi_{sjt})\right)^{\varepsilon}\right]^{\frac{1}{\varepsilon}} dj\right). \tag{71}$$

Because everything in the integrand depends only on the gaps (m, k, n), the expression can be written in discrete state space as

$$Y_t = Q_t \, \omega_t^{-1} \exp\left(\sum_{m=1}^{\bar{m}} \sum_{k=1}^{\bar{k}} \sum_{n=1}^{\bar{n}} \underbrace{\ln\left[(\xi\theta^k)^{\beta}\left(\frac{((1-\xi)\theta^{kt})^{\alpha}}{\gamma n^{\alpha_s}} \frac{\phi_t(m,k,n)}{\sigma_t(m,k,n)}\right)^{\varepsilon} + \left(\lambda^{-mt}(1-\phi_t(m,k,n))\right)^{\varepsilon}\right]^{\frac{1}{\varepsilon}}}_{\equiv R_t(m,k,n)} \mu_t(m,k,n)\right). \tag{72}$$

$$\begin{aligned}\ln Y_{t+\Delta t} - \ln Y_t = & \left(\ln Q_{t+\Delta t} - \ln Q_t\right) + \ln \omega_t - \ln \omega_{t+\Delta t} \\ & + \sum_{m=1}^{\bar{m}} \sum_{k=1}^{\bar{k}} \sum_{n=1}^{\bar{n}} \Big(R_{t+\Delta t}(m,k,n) - R_t(m,k,n)\Big)\Big(\mu_{t+\Delta t}(m,k,n) - \mu_t(m,k,n)\Big) + o(\Delta t).\end{aligned} \tag{73}$$

where

$$\begin{aligned}\ln Q_{t+\Delta t} - \ln Q_t = & \ln \lambda \left[\sum_{m=1}^{\bar{m}} \sum_{k=1}^{\bar{k}} \sum_{n=1}^{\bar{n}} \left(z_t^{Int}(m,k,n) + p_{s \geq s'}^{\text{Ex}} \, z_t^{Ex}(m,k,n) + Z_t^f(m,k,n)\right)\mu_t(m,k,n)\right]\Delta t \\ & + o(\Delta t).\end{aligned} \tag{74}$$

Dividing by Δt and taking the limit $\Delta t \to 0$, the growth rate of the economy is

$$g_t = -g_{\omega,t} + g_{Q,t} + g_{R,t}. \tag{75}$$

In the steady state, the distribution $\mu_t(m,k,n)$ is constant, implying that R_t is constant. Wages grow at the same rate as output, so the real wage remains constant. Therefore, in steady state the growth rate of the economy is determined solely by quality improvements:

$$g = g_Q = \ln\lambda \left[\sum_{m=1}^{\bar{m}} \sum_{k=1}^{\bar{k}} \sum_{n=1}^{\bar{n}} \Big(z^{\mathrm{Int}}(m,k,n) + p^{\mathrm{Ex}}_{s\geq s'}\, z^{\mathrm{Ex}}(m,k,n) + Z^f(m,k,n) \Big) \mu(m,k,n) \right]. \tag{76}$$

B.6. Decomposition of Output

Starting from (71) and factoring out the term $\Xi(e_{st})\Big(\frac{((1-\xi)e_{st})^\alpha}{\gamma\, n_{st}^{\alpha_s}}\Big)$, aggregate output can be written as

$$Y_t = Q_t\, \omega_t^{-1}\, E_t\, \exp\left(\int_0^1 \ln \left[\left(\tfrac{\phi_{sjt}}{\sigma_{sjt}}\right)^\varepsilon + \frac{1}{(\xi e_{st})^\beta} \left(\frac{\gamma\, n_{st}^{\alpha_s}}{((1-\xi)e_{st})^\alpha}\, \lambda^{-m_{jt}} (1-\phi_{sjt}) \right)^\varepsilon \right]^{1/\varepsilon} dj \right). \tag{77}$$

Define the multiplicative factor that collects the factored-out terms as

$$E_t \;=\; \exp\left(\int_0^1 \ln \left[\Xi(e_{st}) \left(\tfrac{((1-\xi)e_{st})^\alpha}{\gamma\, n_{st}^{\alpha_s}} \right)^\varepsilon \right]^{1/\varepsilon} dj \right). \tag{78}$$

Next multiply and divide the integrand by the linear weight $\dfrac{\phi_{sjt}}{\sigma_{sjt}} + (1-\phi_{sjt})$. After this algebraic step I obtain a decomposition that isolates a simple mean term and a residual term:

$$\begin{aligned} Y_t = {} & Q_t\, E_t\, \omega_t^{-1}\, \exp\left(\int_0^1 \ln \left[\tfrac{\phi_{sjt}}{\sigma_{sjt}} + (1-\phi_{sjt}) \right] dj \right) \\ & \times \exp\left(\int_0^1 \ln \left[\frac{ \left(\frac{1}{\gamma n_{st}^{\alpha_s}} \frac{\phi_{sjt}}{\sigma_{sjt}} \right)^\varepsilon + \dfrac{1}{(\xi e_{st})^\beta} \left(\dfrac{1}{((1-\xi)e_{st})^\alpha} \lambda^{-m_{jt}} (1-\phi_{sjt}) \right)^\varepsilon }{ \frac{\phi_{sjt}}{\sigma_{sjt}} + (1-\phi_{sjt}) } \right]^{1/\varepsilon} dj \right). \end{aligned} \tag{79}$$

Finally, using (34) and defining the multiplicative mean term

$$M_t = \frac{\exp\Big(\int_0^1 \ln\left[\frac{\phi_{sjt}}{\sigma_{sjt}} + (1-\phi_{sjt})\right] dj\Big)}{\int_0^1 \left[\frac{\phi_{sjt}}{\sigma_{sjt}} + (1-\phi_{sjt})\right] dj}, \tag{80}$$

the output decomposition can be written as

$$Y_t = Q_t \times E_t \times M_t \times S_t,$$

where

$$S_t = \exp\left(\int_0^1 \ln\left[\frac{\left(\frac{1}{\gamma\, n_{st}^{\alpha_s}}\frac{\phi_{sjt}}{\sigma_{sjt}}\right)^{\varepsilon} + \frac{1}{(\xi e_{st})^{\beta}}\left(\frac{1}{((1-\xi)e_{st})^{\alpha}}\lambda^{-m_{jt}}(1-\phi_{sjt})\right)^{\varepsilon}}{\frac{\phi_{sjt}}{\sigma_{sjt}} + (1-\phi_{sjt})}\right]^{1/\varepsilon} dj\right). \tag{81}$$

B.7. Consumption Equivalence Welfare Measure

On the balanced growth path, consumption grows at rate g, so that $C(t) = C_0 \exp(gt)$.[39] Defining welfare Ω as the present value of lifetime utility from consumption yields:

$$\Omega = \int_0^\infty e^{-\rho t} \ln(C(t))\mathrm{d}t \tag{83}$$

$$= \ln(C_0)\int_0^\infty e^{-\rho t}\mathrm{d}t + g\int_0^\infty te^{-\rho t}\mathrm{d}t. \tag{84}$$

[39]The C_0 consumption level is given by:

$$C_0 = Y_0 - \int_0^1 \left(I_0^{\text{Int}} + I_0^{\text{Emb}} + I_0^{\text{Ex}} + I_0^{f}\right) dj + G_0. \tag{82}$$

In this equation, the subscript 0 represents calibrated optimum values on the balanced growth path, and G is the lump-sum transfer of government taxes.

Solving these integrals gives:

$$\Omega = \frac{\ln(C_0)}{\rho} + \frac{g}{\rho^2} \tag{85}$$

$$= \frac{1}{\rho}\left(\ln C_0 + \frac{g}{\rho}\right). \tag{86}$$

Equivalent Welfare Changes Between Economies To compare welfare between two economies—a calibrated benchmark economy (Cal) and a taxed economy (Tax) on their respective balanced growth paths—I compute the percentage change δ in lifetime consumption that would make households indifferent between the two. The required compensation δ satisfies:

$$\Omega^{\text{Tax}} = \frac{1}{\rho}\left(\ln\left(C_0^{\text{Cal}}(1+\delta)\right) + \frac{g^{\text{Cal}}}{\rho}\right). \tag{87}$$

Solving equation (87) for δ:

$$\frac{\ln C_0^{\text{Tax}}}{\rho} + \frac{g^{\text{Tax}}}{\rho^2} = \frac{\ln[C_0^{\text{Cal}}(1+\delta)]}{\rho} + \frac{g^{\text{Cal}}}{\rho^2} \tag{88}$$

$$\ln\left(\frac{C_0^{\text{Tax}}}{C_0^{\text{Cal}}}\right) + \frac{g^{\text{Tax}} - g^{\text{Cal}}}{\rho} = \ln(1+\delta) \tag{89}$$

$$\delta = \frac{C_0^{\text{Tax}}}{C_0^{\text{Cal}}}\exp\left(\frac{g^{\text{Tax}} - g^{\text{Cal}}}{\rho}\right) - 1. \tag{90}$$

If $\delta > 0$: households require compensation to remain in the benchmark economy (Cal).
If $\delta < 0$: households would pay to move to the taxed economy (Tax).

C. Numerical Appendix

C.1. Additional Numerical Results

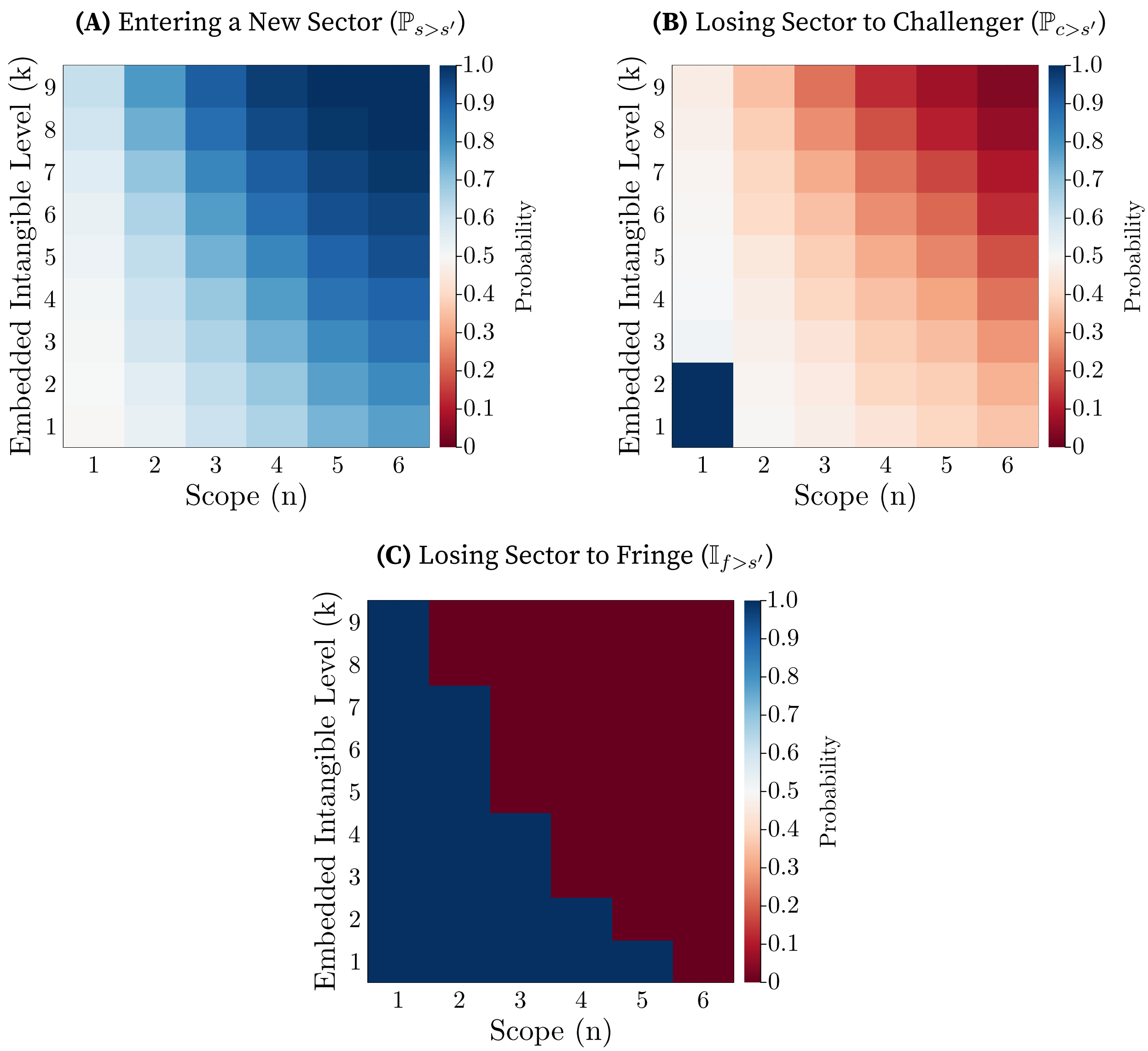


Figure C1. Superstar Firm Entry and Displacement Probabilities

Note: Each panel displays a probability evaluated over the (n, k) grid, where the horizontal axis reports firm scope n and the vertical axis reports the embedded intangible level k. Darker blue indicates higher probability and darker red indicates lower probability.

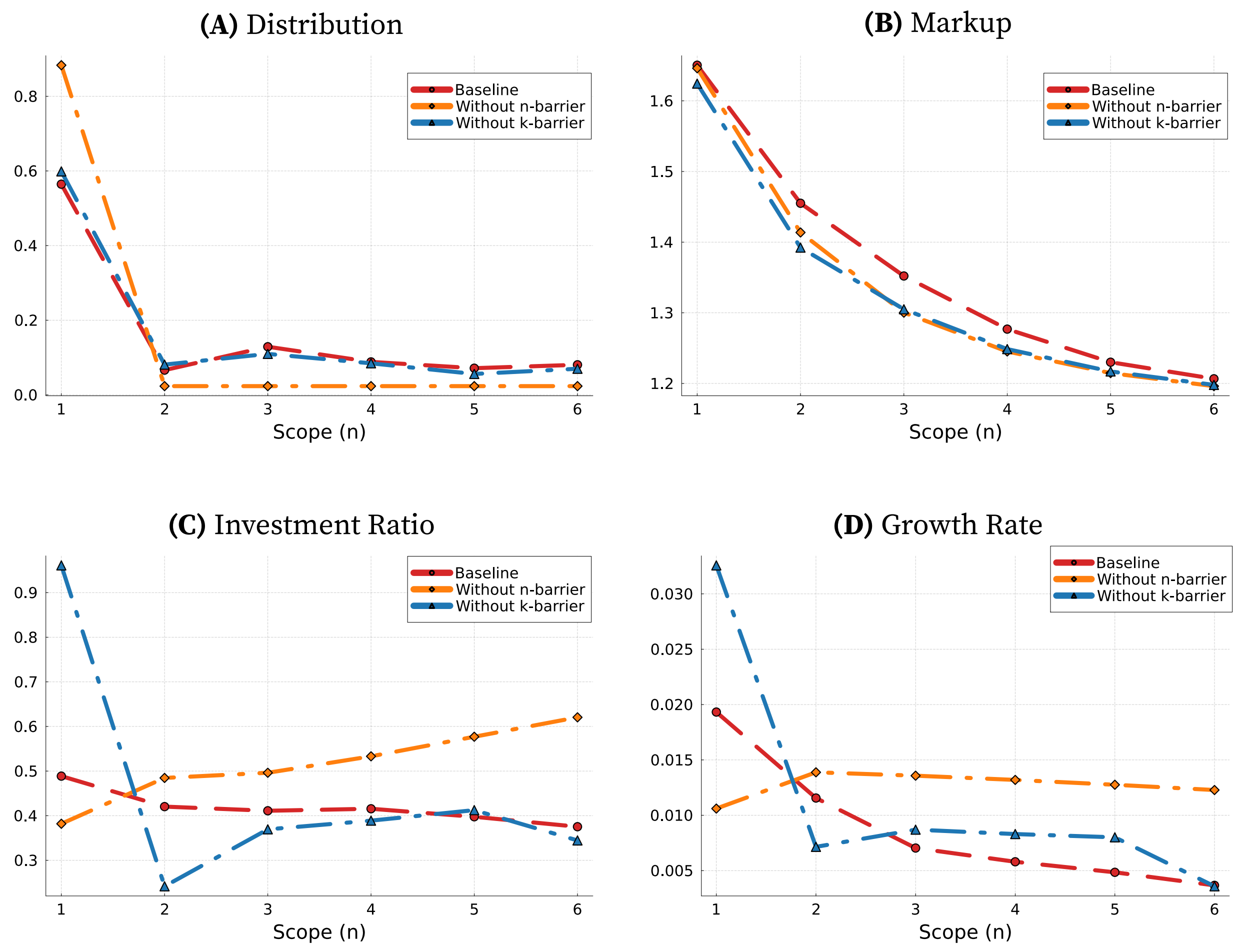
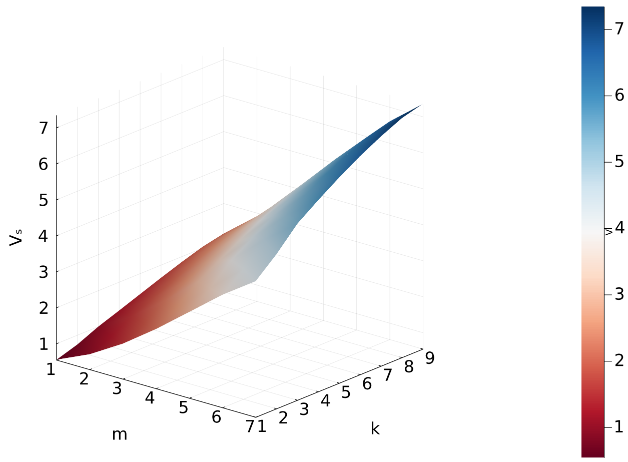


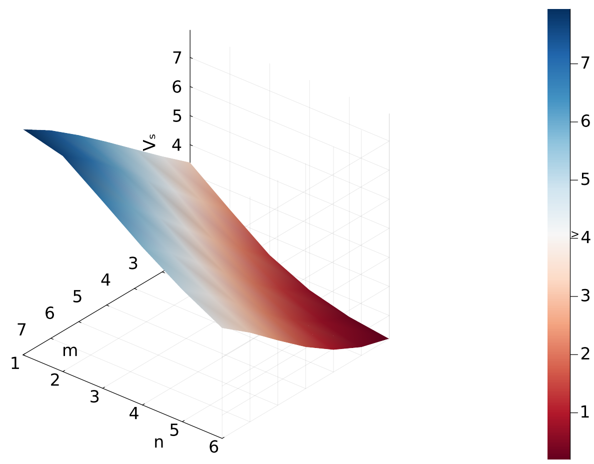
Figure C2. Removing Entry Barrier Scenarios

Note: The green line represents the internally calibrated optimal value of ξ, while the blue line shows an upward shift in ξ and the orange line shows a downward shift. The horizontal axis corresponds to the production line dimension.

(A) $V_s(m,k)$



(B) $V_s(m,n)$



(C) $V_s(k,n)$

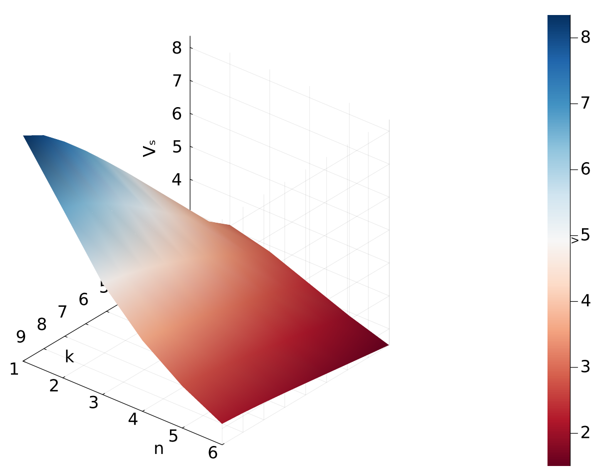


Figure C3. Superstar Value Function

Table C1. Policy Experiments: Growth and Welfare Effects ($|\tau| = 0.2$)

	Without Fringe Subsidy		*With Fringe Subsidy* ($\tau_f = 0.2$)	
Policy Pattern	Δg (%)	$\Delta\Omega$ (%)	Δg (%)	$\Delta\Omega$ (%)
Panel A: Scope Tax (τ_{scope})				
All $+0.2$	-30.886	-2.130	-15.568	5.362
First $1 = -0.2$, rest $+0.2$	-11.455	12.363	0.679	16.333
First $2 = -0.2$, rest $+0.2$	31.544	-7.841	62.335	-7.086
First $3 = -0.2$, rest $+0.2$	36.502	-6.418	67.685	-5.293
First $4 = -0.2$, rest $+0.2$	36.611	-6.360	67.600	-5.113
First $5 = -0.2$, rest $+0.2$	36.770	-6.318	67.541	-5.047
All -0.2	37.592	-6.030	68.384	-4.778
Panel B: Horizontal Innovation Tax (τ_{hor})				
All $+0.2$	-1.186	2.350	21.265	10.739
First $1 = -0.2$, rest $+0.2$	4.114	-2.524	29.601	3.707
First $2 = -0.2$, rest $+0.2$	3.577	-2.505	28.827	3.823
First $3 = -0.2$, rest $+0.2$	2.359	-2.768	27.276	3.572
First $4 = -0.2$, rest $+0.2$	1.311	-3.023	25.925	3.305
First $5 = -0.2$, rest $+0.2$	0.589	-3.184	25.010	3.146
All -0.2	0.589	-3.184	25.010	3.146
Panel C: Embedded Intangible Tax (τ_{emb})				
All $+0.2$	-5.047	1.793	17.426	10.798
First $1 = -0.2$, rest $+0.2$	-2.738	-2.072	20.350	5.155
First $2 = -0.2$, rest $+0.2$	-0.748	-3.030	22.943	3.520
First $3 = -0.2$, rest $+0.2$	1.223	-3.247	25.367	3.091
First $4 = -0.2$, rest $+0.2$	2.949	-3.509	27.497	2.624
First $5 = -0.2$, rest $+0.2$	4.380	-3.793	29.245	2.138
First $6 = -0.2$, rest $+0.2$	5.414	-4.089	30.530	1.682
First $7 = -0.2$, rest $+0.2$	6.180	-4.350	31.527	1.204
First $8 = -0.2$, rest $+0.2$	6.499	-4.492	31.816	1.111
All -0.2	6.499	-4.492	31.816	1.111

Notes: Each row reports the percentage change in the balanced growth rate (Δg) and welfare (Ω) relative to the baseline equilibrium under a tax or subsidy of magnitude $|\tau| = 0.2$. Panel A applies the policy to scope expansion, Panel B to horizontal innovation, and Panel C to embedded intangible investment. The left columns report results without a fringe subsidy; the right columns add a simultaneous fringe subsidy of $\tau_f = 0.2$. "First $j = -0.2$, rest $+0.2$" denotes a subsidy applied to the first j scope levels and a tax on the remaining levels.

Table C2. Sensitivity Analysis: Effect of a $\pm 1\%$ Parameter Perturbation on Model Outcomes

	Investment Ratio		Markup		Growth Rate	
Parameter	$+1\%$	-1%	$+1\%$	-1%	$+1\%$	-1%
Demand and Technology						
CES parameter (ε)	1.960	−1.447	−0.096	0.173	2.779	−2.559
Curvature of managerial quality (α)	1.305	127.472	0.171	−1.769	3.474	4.354
Curvature of brand value (β)	−0.062	0.063	−0.054	0.054	−0.065	0.066
Scale of managerial quality (γ)	−2.932	3.130	−0.423	0.455	−1.602	1.626
Share of brand value (ξ)	−0.586	0.581	−0.085	0.085	−0.287	0.298
Quality improvement step size (λ)	334.219	−23.373	3.839	−4.457	62.250	−34.154
Embedded innovation step size (θ)	14.919	112.033	0.284	−2.206	8.750	1.256
Curvature and Cost Scale Parameters						
Curvature of embedded innovation (ϑ_{Emb})	1.519	−1.532	0.016	−0.015	0.923	−0.935
Cost scale of internal innovation (γ^{Ver})	−0.422	0.431	0.008	−0.008	−0.202	0.206
Cost scale of horizontal innovation (γ^{Hor})	−0.291	0.297	0.032	−0.032	−0.102	0.110
Cost scale of embedded innovation (γ^{Emb})	−0.026	0.026	−0.002	0.002	−0.102	0.104
Cost scale of fringe (γ^{f})	−0.493	0.504	−0.031	0.031	−0.819	0.835
Baseline value	1.3724		1.5731		0.0116	

Notes: Each entry reports the percentage change in the outcome when the row parameter is perturbed by $+1\%$ or -1%, holding all other parameters at their baseline values. The *Intangibility Ratio* is defined as $\left(G_{\text{ver}} + G_{\text{hor}} + G_f\right)/G_{\text{emb}}$, where $G_k = \gamma_k\, z_k^{\vartheta_k}$. The *Markup* is $(1-\varepsilon)/\left[(1-\varphi)\,\varepsilon\right]+1$. The *Growth Rate* is $\mathbb{E}\left[\log \lambda \cdot \left(z_{\text{ver}} + p_{\text{ex}}\, z_{\text{hor}} + \bar{m}\, p_f\, z_f\right)\right]$. Asymmetry between the $+1\%$ and -1% columns reflects local nonlinearity of the equilibrium mapping.

C.2. Solution Algorithm

This section characterizes the numerical algorithm for computing the balanced growth path equilibrium over the three-dimensional state space (m, k, n). The solution involves finding the superstar value functions $v(m, k, n; \mu)$ and fringe value functions $v_f(m, k, n; \mu)$, the innovation rates $z^{\text{Ver}}, z^{\text{Emb}}, z^{\text{Hor}}, z^f$, and the stationary distribution $\mu(m, k, n)$ that jointly satisfy the model's equilibrium conditions. A key feature of the model is that the value functions (equations (26)–(27)) depend directly on the distribution μ (equation (32)), which in turn is determined by the innovation rates defined in equations (28)–(31) that are themselves derived from the value functions. This mutual dependence requires solving the HJB equations and the Kolmogorov Forward Equation (KFE) jointly through nested iteration.

BGP Equilibrium Solution:

1. **Compute static values:** Calculate static market shares and profit values using equations (19) and (23).
2. **Initialization:** Initialize the value functions $v^{(0)}(m, k, n; \mu^{(0)})$ and $v_f^{(0)}(m, k, n; \mu^{(0)})$, and the stationary distribution $\mu^{(0)}(m, k, n)$.
3. **Outer Loop: Iterate over (v, μ) until joint convergence**
   Repeat until $\max |v^{\text{new}} - v^{\text{old}}| < \varepsilon$ and $\max |\mu^{\text{new}} - \mu^{\text{old}}| < \varepsilon$:
   (a) **Step 1: Solve HJB Equations (Inner Loop, given μ^{old})**
      - Set $v^{\text{old}}(m, k, n; \mu^{\text{old}})$.
      - Repeat until $\max |v^{\text{new}} - v^{\text{old}}| < \varepsilon$:
        - Compute policy functions $z^{\text{Ver}}, z^{\text{Emb}}, z^{\text{Hor}}, z^f$ from the first-order conditions using v^{old}.
        - Solve the discretized HJB equations for $v^{\text{new}}(m, k, n; \mu^{\text{old}})$.
        - Update $v^{\text{old}} \leftarrow v^{\text{new}}$.

   (b) **Step 2: Solve the Kolmogorov Forward Equation (Inner Loop, given v^{new})**

- Set $\mu^{\text{old}}(m, k, n; v^{\text{new}})$.
- Repeat until $\max|\mu^{\text{new}} - \mu^{\text{old}}| < \varepsilon$:
  - Solve the KFE for $\mu^{\text{new}}(m, k, n; v^{\text{new}})$ using the policy functions implied by v^{new}.
  - Update $\mu^{\text{old}} \leftarrow \mu^{\text{new}}$.

(c) **Step 3: Update and check outer convergence**

Pass μ^{new} back into the HJB equations, updating $v^{\text{old}} \leftarrow v^{\text{new}}$ and $\mu^{\text{old}} \leftarrow \mu^{\text{new}}$, and repeat Steps 1–2 until both v^{new} and μ^{new} have converged jointly.

(d) **Parameter Estimation:** To find parameters, search over the parameter space to minimize the objective function,

$$\mathcal{L} = \sum_{z=1}^{Z} \frac{\left|\text{model}(z) - \text{data}(z)\right|}{\frac{1}{2}\left|\text{model}(z)\right| + \frac{1}{2}\left|\text{data}(z)\right|},$$

where each evaluation of $\mathcal{L}$ requires a full solution of the BGP equilibrium through the nested iteration described above.